Higher Order Differential 0Equation:

At times the specification of a model may involve the second derivative or a derivative of an even higher order. We may for instance given a function describing the rate of change of a rate of change of the income variable y, say,

π=3.14159.......

From which we are supposed to find the time path of y. In this event the given function constitutes a second order differential equation and the task of finding the time path $y_{(t)}$ is that of solving the second order differential equation

A simple variety of linear differential equation of order n is of the following form.

2π rad

Or in an alternative notation

2π rad

This equation is of the order n because the nth derivative is the highest derivative present. It is linear since all the derivatives as well as dependent variable y appears only in the first degree and no product term occurs in which y and any of its derivatives are multiplied together. Moreover this differential equation is characterized by constant coefficient (a) and term (b).

Second Order Linear Differential Equation with Constant Coefficient and Constant Term:

Suppose we are seeking the solution of 2nd order differential equation, i.e.

2π

Where a_1, a_2, and b are all constant the general solution will consist of the sum of the two components a particular integral y_p and complementary function y_c.

The Particular Integral:

Since the particular integral can be any solution of non homogenous (when b is a non zero constant) or complete equation, we should always try the simplest possible type. Namely y=k (constant). It follows that

$\frac{3\pi}{2}$

Hence the above equation will reduce to

π

If however a_2=0 we must try a solution of the form $\frac{\pi}{2}$ then $\frac{\pi}{4}$ but θ and the above equation can be reduced

θ

Since y_p is non constant, it represents moving equilibrium. If it happens that θ then try a solution of the form θ then θ and θ and the above differential equation can be written as

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θ

Complementary Function:

Since a complementary function is defined to be general solution of the reduced equation (when term b is identically zero), i.e.

$$2\pi \quad rad$$

Trying a solution this form

$$\theta$$

Then

Hence we can rewrite the above reduced equation as

$$2\ \pi$$

Dividing both sides by Ae^{rt}, we get

$Sin(\theta+2n\pi)=Sin\theta \quad Cos(\theta+2n\pi)=Cos\theta$

This equation is known as the characteristic equation or the auxiliary equation. It yields two roots (called characteristic roots) as follows

$$\theta$$

Regarding these roots we have three possible cases.

Case 1(distinct real roots):

When $\frac{1}{2}\pi$, the square root in (i) is a real number and two roots r_1 and r_2 will take distinct real values because the square root is added to $-a_1$ for r_1, but subtracted from $-a_1$, for r_2. In this case the complementary function will be

$$\pi$$

Case 2(repeated real roots)

When $\frac{3}{2}\pi$ the square root in (i) will vanish and the two characteristic roots take an identical value

$$2\ \pi$$

Such roots are known as repeated roots. In this case the complementary function will be written as

$$\sin\theta$$

Case3 (complex roots):

When $\cos\theta$ the characteristic roots will be the pair of conjugate complex numbers

$\sin(-\theta)\equiv-\sin\theta$

Where $\cos(-\theta)\equiv-\cos\theta$ and $\sin^2\theta+\cos^2\theta\equiv1\ [\text{where } \sin^2\theta\equiv(\sin\theta)^2, \text{etc.}]$

The complementary function will thus be in the form.

$\sin(\theta_1\pm\theta_2)\equiv\sin\theta_1\cos\theta_2\pm\cos\theta_1\sin\theta_2$

The general function is the sum of the complementary function y_c and particular integral y_p, i.e.

$\cos(\theta_1\pm\theta_2)\equiv\cos\theta_1\cos\theta_2\mp\sin\theta_1\sin\theta_2$

To define the solution initial condition will be required.

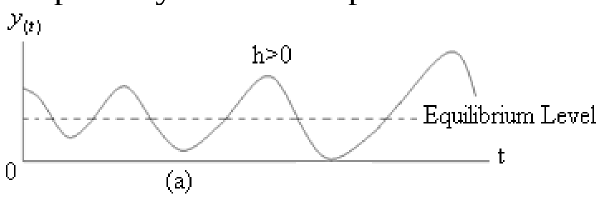
In order to discuss the time path of

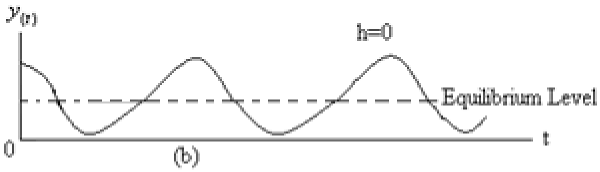
$$\frac{d}{d\theta}\sin\theta=\cos\theta$$

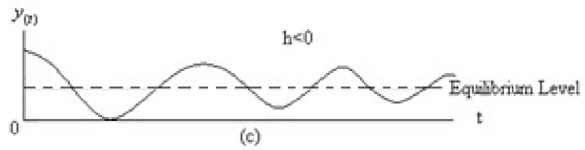
We have to take into account the value of h. if h>0, the value of e^{ht} will increase continually as t increases. This will produce a magnifying effect on the amplitude of ($A_5 \cos vt + A_6 \sin vt$). The time path will in this case be characterized by the “Explosive Fluctuation”

If h=0 on the other hand, then $e^{ht}=1$. Each cycle will display a uniform pattern of deviation from the equilibrium. This is a time path with uniform fluctuation.

If h<0 the term e^{ht} will continually decrease as t increases and the time path is characterized by damped fluctuation. Graphically it can be represented as.







The Dynamic Stability of Equilibrium:

For case 1 and 2, the condition for the dynamic stability of equilibrium again depends upon the algebraic signs of the characteristic roots. For case 1, the complementary function consists of the two exponential expressions $\frac{d}{d\theta}\cos\theta=-\sin\theta$ and $\left.\begin{matrix}\phi'(\theta)=\cos\theta\\ \phi''(\theta)=-\sin\theta\\ \phi'''(\theta)=-\cos\theta\\ \phi^4(\theta)=\sin\theta\\ \phi^5(\theta)=\cos\theta\\ \vdots\quad\vdots\end{matrix}\right\}\Rightarrow\left\{\begin{matrix}\phi'(0)=\cos 0=1\\ \phi''(0)=-\sin 0=0\\ \phi'''(0)=-\cos 0=-1\\ \phi^4(0)=\sin 0=0\\ \phi^5(0)=\cos 0=1\\ \vdots\quad\vdots\end{matrix}\right.$. The coefficients A_1 and A_2 are arbitrary constants; their values hinge on the initial conditions of the problem. Thus we can be sure of a dynamically stable equilibrium $\sin\theta=0+\theta+0-\frac{\theta^3}{3!}+0+\frac{\theta^5}{5!}+......+\frac{\phi^{(n+1)}(p)}{(n+1)!}\theta^{n+1}$, regardless of what initial conditions happen to be, if and only if the roots r_1 and r_2 are both negative. We emphasize the word "both" here, because the condition for the dynamic stability does not permit even one of the roots to be positive or zero. For case 2, with repeated roots the complementary function contains not only the familiar e^{rt} expression but also a multiplicative expression te^{rt}. For the former term to approach zero whatever the initial condition may be, it is necessary−and−sufficient to have r<0

For case 3, specifically the equilibrium is dynamically stable if and only if the time path is convergent. The condition for the convergence of the $y_{(t)}$ path, namely h<0 is therefore also the condition for dynamic stability of the equilibrium of y.

Example 1:

Find the particular integral of the equation.

$$\sin\theta=\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-\frac{\theta^7}{7!}+...........(i)$$

$$\psi(\theta)=\cos\theta$$

Example 2:

Find the y_p of the equation

$$\psi(0)=\cos 0=1$$

Since $a_2=0$

$$\left.\begin{matrix}\psi'(\theta)=-\sin\theta\\ \psi''(\theta)=-\cos\theta\\ \psi'''(\theta)=\sin\theta\\ \psi^4(\theta)=\cos\theta\\ \psi^5(\theta)=-\sin\theta\\ \vdots\quad\vdots\end{matrix}\right\}\Rightarrow\left\{\begin{matrix}\psi'(0)=-\sin 0=0\\ \psi''(0)=-\cos 0=-1\\ \psi'''(0)=\sin 0=-1\\ \psi^4(0)=\cos 0=0\\ \psi^5(0)=-\sin 0=1\\ \vdots\quad\vdots\end{matrix}\right.$$

Example 3:

Find the y_p of the equation $\cos\theta=1+0-\frac{\theta^2}{2!}+0+\frac{\theta^4}{4!}+.......+\frac{\psi^{(n+1)}(p)}{(n+1)!}\theta^{n+1}$ since $\cos\theta=1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-\frac{\theta^6}{6!}+.........(ii)$. Hence we can apply following formula for particular integral

$$e^x=1+x+\frac{1}{2!}x^2+\frac{1}{3!}x^3+\frac{1}{4!}x^4+......$$

Example 4:

Solve the differential equation

$$e^{i\theta}=1+i\theta+\frac{(i\theta)^2}{2!}+\frac{(i\theta)^3}{3!}+\frac{(i\theta)^4}{4!}+\frac{(i\theta)^5}{5!}+........$$

For y_p see example 1. Hence y_p =5

For equation complementary function, the above equation has a characteristic equation, i.e.

$$e^{i\theta}=1+i\theta-\frac{\theta^2}{2!}-\frac{i\theta^3}{3!}+\frac{\theta^4}{4!}+\frac{i\theta^5}{5!}-........$$

$$e^{i\theta}=\left(1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-.....\right)+i\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-......\right)\rightarrow(a)$$

The complementary function will be

$$e^{-i\theta}=1-i\theta+\frac{(-i\theta)^2}{2!}+\frac{(-i\theta)^3}{3!}+\frac{(-i\theta)^4}{4!}+\frac{(-i\theta)^5}{5!}+....$$

Hence the general solution will be

$$e^{-i\theta}=1-i\theta+\frac{\theta^2}{2!}+\frac{i\theta^3}{3!}+\frac{\theta^4}{4!}-\frac{i\theta^5}{5!}-....$$

To definitize solution let t=0

$$e^{-i\theta}=\left(1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-......\right)-i\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-......\right)\rightarrow(b)$$

Differentiating (i) W.r.t “t”

$$e^{i\theta}\equiv\cos\theta+i\sin\theta$$

$$e^{-i\theta}\equiv\cos\theta-i\sin\theta$$

$$\theta$$

$$\frac{\pi}{6}$$

Or

$$\frac{\pi}{4}$$

$$\frac{7\pi}{3}$$

Hence definite solution will be

$$\frac{3\pi}{4}$$

Validity:

$$\sin\theta$$

Substituting into given differential equation

$$\frac{1}{2}$$

$$\frac{1}{\sqrt{2}}\left(\text{¿}\frac{\sqrt{2}}{2}\right)$$

$$\frac{\sqrt{3}}{2}$$

Example 5:

$$\frac{1}{\sqrt{2}}\left(\text{¿}\frac{\sqrt{2}}{2}\right)$$

$$\cos\theta$$

$$\frac{\sqrt{3}}{2}$$

For y_c, finding characteristic roots

$$\frac{1}{\sqrt{2}}\left(\text{¿}\frac{\sqrt{2}}{2}\right)$$

Hence

$$\frac{1}{2}$$

$$\frac{-1}{\sqrt{2}}\left(\text{¿}\frac{-\sqrt{2}}{2}\right)$$

Set t=0, to definitize the solution

As $r^2+r+4=0 \quad a_1=1,\ a_2=4$

$$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-1\pm\sqrt{(-1)^2-4(4)}}{2}$$
$$=\frac{-1\pm\sqrt{-15}}{2}=\frac{-1\pm\sqrt{15}\sqrt{-1}}{2}=\frac{-1}{2}\pm\frac{\sqrt{15}}{2}i$$
$$(\sqrt{-1}=i)$$

$$r_1+r_2=-a_1$$

Differentiating (i) W.r.t “t”

$$r_1r_2=a_2$$

$$\frac{d\sin\theta}{d\theta}=\cos\theta$$

$$\frac{d^2}{d\theta^2}\sin\theta=\frac{d}{d\theta}\cos\theta=-\sin\theta$$

$$e^{i\pi}=\cos\pi+i\sin\pi$$

$$\cos\pi=-1 \quad and \quad \sin\pi=0$$

$$e^{i\pi}=-1+i\times0$$

$$e^{i\pi}=-1$$

Exercise 15.1:

Q.1. Find the particular integral of each equation

(a) $e^{-i\pi/2}=-i$

$$\theta=\frac{\pi}{2}$$

(b) $e^{-i\pi/2}=\cos\frac{\pi}{2}-i\sin\frac{\pi}{2}=0-i(1)=-i$

$$5e^{3i\pi/2}$$

$$R=5 \quad and \quad \theta=\frac{3\pi}{2}$$

(c)
$$h=R\cos\theta \quad and \quad v=R\sin\theta$$
$$h=5\cos\frac{3\pi}{2} \qquad v=5\sin\frac{3\pi}{2}$$
$$h=5(0)=0 \qquad v=5(-1)=-5$$

$$\cos3\frac{\pi}{2}=0$$
$$\sin3\frac{\pi}{2}=-1$$

$$h+vi=-5i \quad as\ (h=0)$$

(d) $(1+\sqrt{3}\,i)$

$$h=1,\ v=\sqrt{3}$$

$$R=\sqrt{h^2+v^2}=\sqrt{(1)^2+(\sqrt{3})^2}=\sqrt{4}=2$$

(e)

Alternative method

(a) Try a solution of the form y=k

(b) Try a solution of the form y=kt

(c) Try a solution of the form y=k

(d) Try a solution of the form y=k

(e) Try a solution of the form $y=kt^2$

Q.2. Find the complementary function of each equation

(a) $h=R\cos\theta \qquad v=R\sin\theta$

$\frac{h}{R}=\cos\theta \qquad \sin\theta=\frac{v}{R}$

$\theta=\cos^{-1}\left(\frac{h}{R}\right) \qquad \theta=\sin^{-1}\left(\frac{v}{R}\right)$

Finding the characteristic roots

$$\theta=\cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3} \qquad \theta=\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{3}$$

$$R(\cos\theta+i\sin\theta)=2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)$$

(b) $Re^{i\theta}=2e^{i\pi/3}$

$$r^2-3r+9=0$$

Finding the characteristic roots

$$a=1,\quad b=-3,\quad c=9$$

$$r_1,\ r_2=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$r_1,\ r_2=\frac{-(-3)\pm\sqrt{(-3)^2-4(1)(9)}}{2(1)}$$

(c) $r_1,\ r_2=\frac{3\pm\sqrt{9-36}}{2}$

$$r_1,\ r_2=\frac{3\pm\sqrt{-27}}{2}=\frac{3\pm\sqrt{27}\sqrt{-1}}{2}$$

$$r_1,\ r_2=\frac{3}{2}\pm\frac{3\sqrt{3}}{2}i$$

$$r^2+2r+17=0$$

$$a=1,\quad b=2,\quad c=17$$

$$r_1,\ r_2=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

(d) $r_1,\ r_2=\frac{-2\pm\sqrt{(2)^2-4(1)(17)}}{2(1)}$

$$r_1,\ r_2=\frac{-2\pm\sqrt{4-68}}{2}$$

Since $r_1,\ r_2=\frac{-2\pm\sqrt{-64}}{2}$

$$r_1,\ r_2=\frac{-2\pm\sqrt{64}\sqrt{-1}}{2}$$

Hence

$$r_1,\ r_2=\frac{-2}{2}\pm\frac{8}{2}i$$

$$r_1,\ r_2=-1\pm\frac{8}{2}i=-1\pm4i$$

Q.3.Find general solution of each differential equation in the preceding problem and then definitize the solution with the initial condition. $2x^2+x+8=0$

(a) $a=2,\quad b=1,\quad c=8$

$r_1,\ r_2=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$$r_1,\ r_2=\frac{-1\pm\sqrt{(1)^2-4(2)(8)}}{2(2)}$$

$$r_1,\ r_2=\frac{-1\pm\sqrt{1-64}}{4}$$

And

$$r_1,\ r_2=\frac{-1\pm\sqrt{-63}}{4}$$

Hence general solution will be

$$r_1,\ r_2=\frac{-1\pm\sqrt{63}\ i}{4}$$

Taking derivative W.r.t‘t’ of equation (i)

$$r_1,\ r_2=\frac{-1\pm3\sqrt{7}\ i}{4}$$

$$r_1,\ r_2=\frac{-1}{4}\pm\frac{3\sqrt{7}}{4}i$$

$$2x^2-x+1=0$$

$a=2,\quad b=-1,\quad c=1$

$r_1,\ r_2=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$$r_1,\ r_2=\frac{-(-1)\pm\sqrt{(-1)^2-4(2)(1)}}{2(2)}$$

(b) $r_1, r_2=\frac{1}{4}\pm\frac{\sqrt{-7}}{4}=\frac{1}{4}\pm\frac{\sqrt{7}\,i}{4}$

$r_1,\ r_2=\frac{1}{4}\pm\frac{\sqrt{7}}{4}i$

$$\pi\ rad=180^0$$

$$rad=\frac{180^0}{\pi}$$

Taking derivative W.r.t "t"

$\pi\ rad=180^0$

Set t=0 to definitize the solution

$\frac{\pi\ rad}{180}=1^o$

$1^o=\frac{\pi\ rad}{180}$

$$a'$$

$\sin^2\theta+\cos^2\theta=1$

$R^2=h^2+v^2 \quad h=R\cos\theta$

$v=R\ \sin\theta$

$R^2=(R\ \cos\theta)^2+(R\ \sin\theta)^2$

(c) $R^2=R^2\ \cos^2\theta+R^2\ \sin^2\theta$

$R^2=R^2\ (\ \cos^2\theta+\ \sin^2\theta)$

$\frac{R^2}{R^2}=\cos^2\theta+\sin^2\theta$

$$1=\cos^2\theta+\sin^2\theta$$

$$\sin\frac{\pi}{4}=\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$$

Taking derivative W.r.t'ť' of equation (i)

$\sin\left(\pi-\frac{3}{4}\pi\right)=\sin\pi\ \cos\frac{3\pi}{4}-\cos\pi\ \sin\frac{3\pi}{4}$

$$\sin\left(\pi-\frac{3}{4}\pi\right)=0-(-1)\frac{1}{\sqrt{2}}$$

Hence

$$\sin\left(\pi-\frac{3}{4}\pi\right)=\frac{1}{\sqrt{2}}$$

(d) $\cos\left(\pi-\frac{3}{4}\pi\right)=\cos\pi\cos\frac{3}{4}\pi+\sin\pi\sin\frac{3}{4}\pi$

Since it is a homogenous equation y_p cannot be calculated. Hence the general solution will be

$\cos\left(\pi-\frac{3}{4}\pi\right)=(-1)\left(-\frac{1}{\sqrt{2}}\right)+0$

Since

$$\cos\left(\pi-\frac{3}{4}\pi\right)=\frac{1}{\sqrt{2}}$$

Definitizing the solution

$$\sin\frac{\pi}{4}=\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$$

$$\sin 2\theta=2\sin\theta\cos\theta$$

$\sin(\theta+\theta)=\sin\theta\cos\theta+\cos\theta\sin\theta$

$$\sin(\theta+\theta)=2\sin\theta\cos\theta$$

Q.4. Are the inter temporal equilibrium found in preceding problem dynamically stable.

(a) $\cos 2\theta=1-\sin^2\theta$

Since both roots are not negative.

Therefore as $\cos(\theta+\theta)=\cos\theta\cos\theta-\sin\theta\sin\theta\ \cos(\theta+\theta)=\cos^2\theta-\sin^2\theta$ approaches to $\cos(\theta+\theta)=[1-\sin^2\theta]-\sin^2\theta$

Hence equilibrium will be dynamically unstable and time path will be divergent.

(b) $\cos(\theta+\theta)=1-2\sin^2\theta$

Since both roots are negative therefore as $\sin(\theta_1+\theta_2)+\sin(\theta_1-\theta_2)=2\sin\theta_1\cos\theta_2$ $\sin\theta_1\cos\theta_2+\cos\theta_1\sin\theta_2+\sin\theta_1\cos\theta_2-\cos\theta_1\sin\theta_2=2\sin\theta_1\cos\theta_2$ Hence inter temporal equilibrium will be dynamically stable.

(c) $1+\tan^2\theta=\frac{1}{\cos^2\theta}$

As $1+\frac{\sin^2\theta}{\cos^2\theta}=\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}$ As $\cos^2\theta+\sin^2\theta=1$ equilibrium is dynamically unstable and time path is divergent.

(d) $=\frac{1}{\cos^2\theta}$

$L.H.S = R.H.S \quad \sin\left(\frac{\pi}{2}-\theta\right)=\cos\theta$. Hence inter temporal equilibrium is dynamically stable and time path is convergent.

Q.5. Verify that the definite solution in example 5 indeed (a) satisfies the two initial conditions and (b) has first and second derivative that confirm to the given differential equation.

Definite solution in example 5 is

$\sin\frac{\pi}{2}\cos\theta-\cos\frac{\pi}{2}\sin\theta=\cos\theta$

(a) $\frac{d^2Y}{dt^2}=kY$

Taking derivative W.r.t'‘t’ of equation (i)

$\frac{d^n y}{dt^n}+a_1\frac{d^{n-1}y}{dt^{n-1}}+\dots\dots\dots+a_{n-1}\frac{dy}{dt}+a_n y=b$

$y^{(n)}(t)+a_1 y^{(n-1)}(t)+\dots..a_{n-1}y'_{(t)}+a_n y=b$

$y''_{(t)}+a_1 y'_{(t)}+a_2 y=b$

$y'_{(t)}=y''_{(t)}=0$ Hence definite solution satisfies the two initial conditions.

(b) $a_2 k=b$

$k=\frac{b}{a_2}$

$y_p=\frac{b}{a_2} \qquad (a_2\neq 0)$

$$y=kt$$

$$y'=k$$

$$y''=0$$

$a_1 k=b$

$k=\frac{b}{a_1}$

$y_p=kt=\frac{b}{a_1}t \qquad (a_2=0, \quad a_1\neq 0)$

$(a_2=a_1=0)$

Q.6. Show that as $y=kt^2$ the limits of te[rt] is zero. If $y'_{(t)}=2kt$ but is infinite if $y''_{(t)}=2k$

$2k=b$

$k=\frac{b}{2}$

(i) $y_p=kt^2=\frac{b}{2}t^2 \qquad (a_1=a_2=0)$

Since denominator and numerator approaches to infinity

Applying L Hospital rule

$$y''_{(t)}+a_1 y'_{(t)}+a_2 y=0$$

(ii)If $y=Ae^{rt}$

$y'=rAe^{rt} \qquad (Ae^{rt}\neq 0)$

$y''=r^2Ae^{rt}$

$r^2Ae^{rt}+a_1 rAe^{rt}+a_2 Ae^{rt}=0$

$Ae^{rt}(r^2+a_1 r+a_2)=0$

Imaginary and Complex Number:

Conceptually it is possible to define a number $r^2+a_1 r+a_2=0$ which when squared will equal −1. Because $\frac{a_1}{2}$ is square root of a negative number, it is obviously not real valued; it is therefore referred to as an imaginary number. Such as

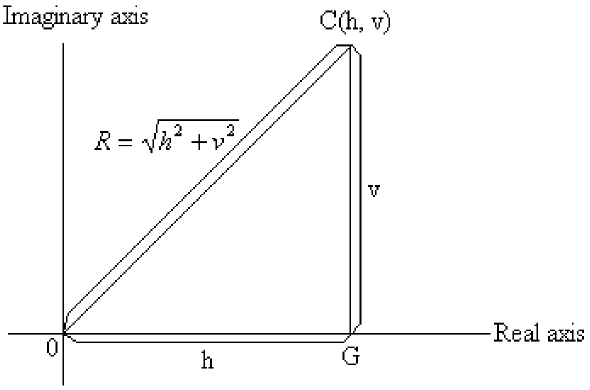
$$a_1^2>4a_2$$

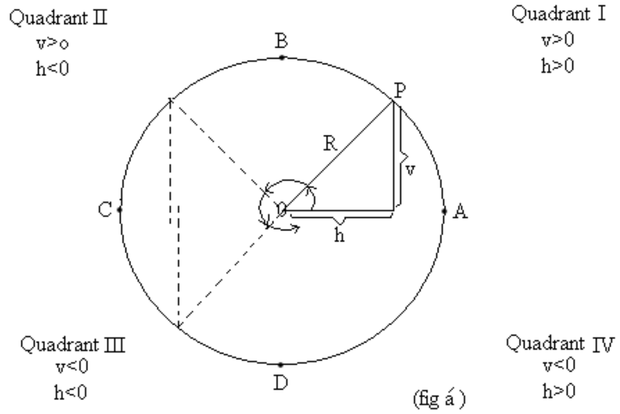
We may construct yet another type of number one that contains a real part as well as imaginary part such as $y_c=Ae^{r_1 t}+A_2 e^{r_2 t} \quad (r_1 \neq r_2)$ known as complex numbers these can be represented generally in the form $a_1^2=4a_2$ where h and v are two real numbers. Of course in case v=0, the complex number will reduce to a real number whereas if h=0 it will become an imaginary number. Thus the set of all real numbers (call it R) constitutes a subset of the set of all complex numbers (call it c). Similarly, the set of all imaginary numbers (call it I) also constitutes a subset of c, i.e.

$r(=r_1=r_2)=-\frac{a_1}{2}$

Argand Diagram:

A complex number (h+vi) can be represented graphically in what is called Argand diagram as illustrated in following figure. By plotting h horizontally on the real axis and v vertically on the imaginary axis, the number (h+vi) can be specified by the point (h, v) which we have alternatively labeled c.





Given the values of h and v, we can also calculate the length of line OC by applying Pythagoras theorem, which states that the square of the hypotenuse of a right−angled triangle is the sum of the squares of the other two sides. Denoting the length of OC by R(for radius vector) we have

$$y_c = A_3 e^{rt} + A_4 t e^{rt}$$

Circular Functions:

Consider a circle with its center at the point of origin and with the radius of length R as shown in figure. Let the radius like the hand of a clock rotate in the counter clock wise direction. Starting from the position OA it will gradually move into the position OP followed successively by such positions as OB, OC and OD; and at the end of a cycle it will return to OA. Therefore the cycle will simply repeat itself.

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When in a specific position−say, OP−the clock hand will make a definite angle with the line OA and the tip of the hand (P) will determine a vertical distance V and a horizontal distance h. As the angle changes during the process of rotation, v and h will vary, although R will not. Thus the ratios v/R and h/R must change with ; that is these two ratios are both functions of the angle . Specifically, v/R and h/R are called respectively, the sin (function) of and the cosine (function) of :

$$y_c = e^{ht}(A_3 \cos vt + A_4 \sin vt)$$

In the view of these connections with a circle these functions are referred to as circular function.

Radian:

Usually angles are measured in degrees (for example, 30^0, 45^0 and 90^0); in analytical work, however, it is more convenient to measure angles in radians instead. But just how much is a radian? To explain this let us return to above figure(a) where we have drawn the point P so that the length of the arc AP is exactly equal to the radius R. A radian (abbreviated as rad) can then be defined as the size of the angle in (fig) formed by such an R−length arc. Since the circumference of the circle has a total length of $y_c \to 0$ As $t \to \infty$ (where $y''_{(t)}+y'_{(t)}-2y=-10$ $a_1=1$ $a_2=-2$ $b=-10$), a complete circle must involve an angle of $y_p=\frac{b}{a_2}=\frac{-10}{-2}=5$ altogether. In terms of degrees, however a complete circle makes an angle of 360^0; thus by equating 360^0 to $y''_{(t)}+y'_{(t)}=-10$ $a_1=1$ $a_2=0$ $b=-10$, we can arrive at the following conversion table:

Degrees	360	270	180	90	45	0
Radians	$y_p=\frac{b}{a_1}t=-10t$	$y''_{(t)}=-10$	$a_1=0$ $a_2=0$	[illegible]	[illegible]	0

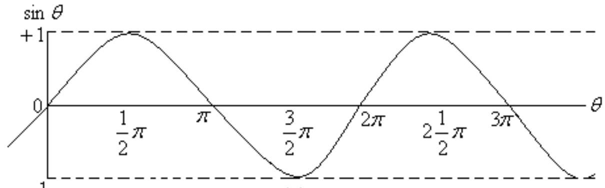
Properties of sin an cos functions:

1. Given the length of R, the value of sin [illegible] hinges upon the way the value of v changes in response to changes in the angle [illegible]. In the starting position OA, we have v=0.
2. The value of cos [illegible], in contrast, depends instead upon the way that h changes in response to changes in [illegible].
3. The sin [illegible] and cos [illegible] functions share the same domain, namely the set of all real numbers (radian measures of [illegible])
4. A major distinguishing property of the sine and cosine functions is that both are periodic; their values will repeat themselves for every $A_1+A_2+5=12$ $A_1-2A_2=-2$ (a complete circle) the angle $A_1+A_2=7$ $\pm A_1 \mp 2A_2=\mp 2$ $3A_2=9$ travels through. Each function is said to have a period of $3A_2=9$ $A_2=\frac{9}{3}=3$ $A_2=3$ $A_1+A_2=7$ $A_1=7-A_2=7-3=4$ $A_1=4$. In view of this periodically feature, the following equation hold (for n any integer):

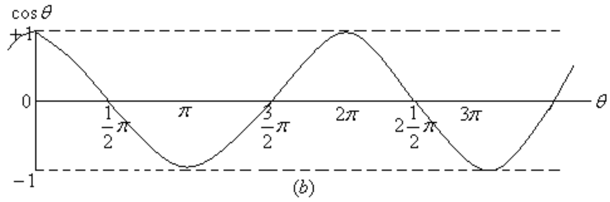
$$y_{(t)}=4e^{t}+3e^{-2t}+5$$

Table

[illegible]	0	$y''_{(t)}+y'_{(t)}-2y_{(t)}=-10$	[illegible]	$-10=-10$	$y''_{(t)}+6y'_{(t)}+9y=27$
$a_1=6$ $a_2=9$ $b=27$	0	1	0	−1	0
$y_p=\frac{b}{a_2}=\frac{27}{9}=3$	1	0	−1	0	1



(a)



(b)

5. The graph of the sine and cosine functions indicate a constant range of fluctuations in each period namely ±1.
6. The sine and cosine functions obey certain identities. Among these the move frequently used is

$$r_1, r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-6\pm\sqrt{(6)^2-4(9)}}{2}$$
$$=\frac{-6\pm\sqrt{36-36}}{2}=\frac{-6}{2}=-3$$
$$r_1, r_2=-3$$
$$y_c=A_3e^{rt}+A_4te^{rt}$$
$$y_c=A_3e^{-3t}+A_4te^{-3t}$$
$$y_t=y_c+y_p=A_3e^{-3t}+A_4te^{-3t}+3 \rightarrow (i)$$
General solution
$$y_{(o)}=5$$
$$y'_{(o)}=-5$$

7. Finally a word about derivatives. Being continuous and smooth both sinθ and cosθ are differentiable. The derivatives, d (sinθ)/dθ and d (cosθ)/dθ are obtainable by taking the limits, respectively, of the difference quotients Δ (sinθ)/Δ θ and Δ (cosθ)/Δ θ as Δθ → 0. The results, stated here without proof.

$$y_{(0)}=A_3+3=5 \Rightarrow A_3=2$$
$$y'_{(t)}=-3A_3e^{-3t}-3A_4te^{-3t}+A_4e^{-3t}$$

Euler Relations:

Any functions which have finite, continuous derivatives up to the desired order can be expended into a polynomial function. Moreover if the remainder term R_n in the resulting Taylor series (expansion at any point x_0) or maclaurin series (expansion at $x_0 = 0$) happens to approach zero as the number of terms n because infinite, the polynomial may be written as an infinite series. We shall now expand the sine and cosine functions and then attempt to show how the imaginary exponential expressions can be transformed into the circular functions having equivalent expansions. For the sin function, write φ (θ) =sinθ; it than follows that φ (0) =sin0=0. By successive derivation, we can get

$$y'_{(0)}=-3A_3+A_4=5$$

We can expand the sin function as maclaurin series with remainder term:

$$y'_{(0)}=-3(2)+A_4=-5$$

Now the expression $\varphi^{(n+1)}(p)$ in the last (remainder) term, which represents the (n+1)st derivative evaluated at θ=p, can only take the form of ±cos p or ±sin p and, as such, can only take a value in the interval [−1, 1] regardless of how large n is. On the other hand, (n+1)! will grow rapidly as $n\rightarrow\infty$ in fact much more rapidly than θ^{n+1} as n increases. Hence the remainder term will approach zero as $n\rightarrow\infty$ and we can therefore express the maclaurin series as on infinite series:

$$y'_{(0)}=-6+A_4=-5$$

Similarly if we write $A_4=-5+6=1$ then $A_4=1$ and the successive derivatives will be

$$y_{(t)}=2e^{-3t}+te^{-3t}+3$$

On the basis of these derivatives, we can expand cosθ as follows

$$y''_{(t)}-2y'_{(t)}+5y=2$$

Since the remainder term will again tend towards zero as $n\rightarrow\infty$, the cosine function is also expressible as an infinite series, as follows:

$$a_2=5,\quad b=2$$

$$y_p=\frac{b}{a_2}=\frac{2}{5}$$

Our immediate interest lies in finding the relationship b/w imaginary exponential expressions and circular functions. To this end let us now expand the two exponential expressions $e^{i\theta}$ and $e^{-i\theta}$. As we know

$$y''_{(t)}+y'_{(t)}=7$$

Letting x=*i*θ, therefore we can immediately obtain

$$a_1=1,\quad a_2=0,\quad b=7$$

$$y_p=\frac{b}{a_1}t=\frac{7}{1}t=7t$$

$$y''_{(t)}+3y=9$$

Similarly by setting x=−*i*θ the following result will emerge

$$a_1=0,\quad a_2=3\quad b=9$$

$$y_p=\frac{b}{a_2}=\frac{9}{3}=3$$

$$y''_{(t)}+2y'_{(t)}-y=-4$$

By substituting (i) and (ii) in above 2 results (a) and (b), the following pair of identities known as Euler relations can readily be established:

$a_1=2,\quad a_2=-1,\quad b=-4$

$y_p=\frac{b}{a_2}=\frac{-4}{-1}=4$

Table b:

$y''_{(t)}=12$ $a_1=a_2=0,\ b=12$ $y_p=\frac{b}{2}t^2=\frac{12}{2}t^2=6t^2$	$y''_{(t)}+3y'_{(t)}-4y=12$	$a_1=3,\ a_2=-4,\ b=12$	$r_1,r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-3\pm\sqrt{9-4(-4)}}{2}$ $=\frac{-3\pm5}{2}=-4,1$	$y_c=A_1e^{r_1t}+A_2e^{r_2t}$ $y_c=A_1e^{-4t}+A_2e^{t}$
$y''_{(t)}+6y'_{(t)}+5y=10$	$a_1=6,\ a_2=5,\ b=10$	$r_1,r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-6\pm\sqrt{(6)^2-4(5)}}{2}$ $=\frac{-6\pm4}{2}=-5,-1$	$y_c=A_1e^{r_1t}+A_2e^{r_2t}$ $y_c=A_1e^{-5t}+Ae^{-t}$	$y''_{(t)}-2y'_{(t)}+y=3$
$a_1=-2,\ a_2=1,\ b=3$	$r_1,r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{2\pm\sqrt{(-2)^2-4(1)}}{2}$ $=\frac{2}{2}=1$	$r(=r_1=r_2)=1$	$y_c=A_1e^{rt}+A_2te^{rt}$ $y_c=A_1e^{t}+A_2te^{t}$	$y''_{(t)}+8y'_{(t)}+16y=0$

Example 1:

Find the roots of the characteristic equation

$a_1=8,\quad a_2=16,$

$$a_1^2=4a_2$$

Note:

Check $\begin{matrix}(8)^2=4(16)\\ 64=64\end{matrix}$

$r=r_1=r_2=\frac{-a_1}{2}=\frac{-8}{2}=-4$

Example 2:

Find the slope of the sinθ curve at=π/2.

The slope of the sin curve is given by its derivative (=cosθ) thus, at θ=π/2, the slope should be cos π/2=0

Example 3:

Find the 2nd derivative of sinθ

$y_c=A_1e^{rt}+A_2te^{rt}$

Since $y_c=A_1e^{-4t}+A_2te^{-4t}$

Therefore the 2nd derivative is

$y_{(0)}=4,\quad y'_{(0)}=2$

Example 4:

Find the value of $e^{i\pi}$

By setting θ=π it is found from Euler Relation that

$y''_{(t)}+3y'_{(t)}-4y=12$

Since

$a_1=3,\quad a_2=-4,\quad b=12$

$y_p=\frac{b}{a_2}=\frac{12}{-4}=-3$

$y_c=A_1e^{-4t}+A_2e^{t}$

Example 5:

Show that $\begin{matrix}y_t=y_c=+y_p\\ y_t=A_1e^{-4t}+A_2e^{t}-3\ \to(i)\\ \text{General solution}\end{matrix}$

$y'_t=-4A_1e^{-4t}+A_2e^{t}\ \to(ii)$

Setting $\text{set } t=0 \text{ for equation } (i)$

$y_{(0)}=A_1+A_2-3=4$

$Y'_{(0)}=-4A_1+A_2=2$

Example 6:

Find the Cartesian from of the complex number $\begin{matrix}A_1+A_2=7\\ \mp4A_1\pm A_2=\pm2\\ \hline 5A_1\quad=5\end{matrix}$

Here we have

$A_1=\frac{5}{5}=1$

$A_1+A_2=7$

$A_2=7-A_1=7-1=6$

$y_t=e^{-4t}+6e^{t}-3$

$y''_{(t)}+6y'_{(t)}+5y=10$

Cartesian form

$a_1=6,\quad a_2=5,\quad b=10$

Example 7:

Find the polar and exponential form of

$y_p=\frac{b}{a_2}=\frac{10}{5}=2$
$y_c=A_1e^{-5t}+A_2e^{-t}$
$y_t=y_c+y_p$
$y_t=A_1e^{-5t}+A_2e^{-t}+2\ \to(i)$
General solution

We have $y'_{(t)}=-5A_1e^{-5t}-A_2e^{-t}\ \to(ii)$

Hence

$y_{(0)}=A_1+A_2+2=4$

$y'_{(0)}=-5A_1-A_2=2$

$$\begin{array}{l} A_1+A_2=2\\ -5A_1-A_2=2\\ \hline -4A_1\quad=4\end{array}$$

$$A_1=4/-4=-1$$

$$A_1+A_2=2$$

$$-1+A_2=2$$

Polar form

$$A_2 = 2 + 1 = 3$$

Exponential form

$y_t = -e^{-5t} + 3e^{-t} + 2$

Exercise 15.2:

Q.1. Find the roots of following quadratic equations:

(a) $y''_{(t)} - 2y'_{(t)} + y = 3$

$$a_1 = -2, \quad a_2 = 1, \quad b = 3$$

$$y_p = \frac{b}{a_2} = \frac{3}{1} = 3$$

$y_c = A_1 e^t + A_2 t e^t$

$y_t = y_c + y_p$

$y_t = A_1 e^t + A_2 t e^t + 3 \quad \rightarrow (i)$

General solution

$y'_{(t)} = A_1 e^t + A_2 t e^t + A_2 e^t$

$set \quad t = 0$

(b) $y_{(0)} = A_1 + 3 = 4$

$y'_{(0)} = A_1 + A_2 = 2$

$A_1 = 4 - 3 = 1$

$A_2 = 2 - A_1 = 2 - 1 = 1$

$y_t = e^t + t e^t + 3$

Definitize solution

Or Definite solution

$$y''_{(t)} + 8y'_{(t)} + 16y = 0$$

$$y_{(t)} = y_c + y_p$$

$y_c = A_1 e^{-4t} + A_2 t e^{-4t}$

$y_t = A_1 e^{-4t} + A_2 t e^{-4t}$

General solution

$y'_{(t)} = -4A_1 e^{-4t} - 4A_2 t e^{-4t} + A_2 e^{-4t}$

$set \quad t = 0$

$y_{(0)} = A_1 = 4$

$y'_{(0)} = -4A_1 + A_2 = 2$

$A_1 = 4$

$$A_2 = 2 + 4A_1$$

$A_2 = 2 + 4(4) = 18$

(c) $y_t = 4e^{-4t} + 18te^{-4t}$

Definite solution

$$r_1 = -4 \quad r_2 = 1$$

$$t \rightarrow \infty$$

$$e^{r_2 t}$$

$$\infty$$

$$r_1 = -5 \quad r_2 - 1$$

$$t \rightarrow \infty$$

$$y_c \rightarrow 0$$

(d) $r = 1$

$$t \rightarrow \infty$$

$$y_c \rightarrow \infty$$

$$r = -4$$

$$t \rightarrow \infty$$

Q.2. (a) how many degrees are there in a radian?

$$y_c \rightarrow 0$$

(b) $y_{(t)} = 2e^{-3t} + te^{-3t} + 3 \rightarrow (i)$

$$y_{(0)} = 5 \quad y'_{(0)} = -5$$

Q.3.With reference to figure 15.3, i.e. and by using Pythagoras' theorem, prove that

(a) $set \quad t = 0 \; in \; eq \, (i) \; and \; (ii)$

$$y_{(0)} = 2 + 3 = 5$$

$y'_{(0)} = -6 + 1 = -5$

$y_{(t)} = 2e^{-3t} + te^{-3t} + 3$

$y'_{(t)} = -6e^{-3t} + e^{-3t} - 3te^{-3t}$

$y''_{(t)} = 18e^{-3t} - 3e^{-3t} + 9te^{-3t} - 3e^{-3t}$

$$y''_{(t)} + 6y'_{(t)} + 9y = 27$$

$18e^{-3t}-3e^{-3t}+9te^{-3t}-3e^{-3t}-36e^{-3t}+6e^{-3t}$
$-18te^{-3t}+18e^{-3t}+9te^{-3t}+27=27$

(b) $27=27$

$t \to \infty$

$r<0$

$r \geq 0$

$\lim_{t \to \infty} te^{rt} = \lim_{t \to \infty} \frac{t}{e^{rt}}$

$\lim_{t\to\infty}\frac{t}{e^{rt}}=\lim_{t\to\infty}\frac{d}{dt}\left(\frac{t}{e^{rt}}\right)=\lim_{t\to\infty}\frac{1}{re^{rt}}=\frac{1}{\infty}=0$

$r \geq 0$

Hence

$\lim_{t\to\infty} te^{rt}=re^{\infty}=r(\infty)=\infty$

Q.4. By means of identities show that

(a) $i \equiv \sqrt{-1}$

$\sqrt{-9}=\sqrt{9}\sqrt{-1}=3i \text{ and } \sqrt{-2}=\sqrt{2}\sqrt{-1}=\sqrt{2}\,i$

(b) $(8+i)$ and $(3+5i)$

$(h+vi)$

$R\subset C$ and $I \subset C$

$R^2=h^2+v^2$ and $R=\sqrt{h^2+v^2}$

θ

(c) θ

θ

(d) θ θ

θ

Therefore

$\sin\theta \equiv \frac{v}{R}$

$\cos\theta \equiv \frac{h}{R}$

Hence θ

(e) a'

$2\pi R$

$\pi=3.14159.......$

2π rad

Hence 2π rad

(f) 2π

$\frac{3\pi}{2}$

π

$\frac{\pi}{2}$

Hence $\frac{\pi}{4}$

Q.5. by applying the chain rules

(a) Write out the derivative formula for

θ $\sin\theta$ and θ $\cos\theta$ where θ a function of is θ

(i) θ

(ii) 2π rad

(b) Find the following derivatives

(i) θ

2π

$\sin(\theta+2n\pi)=\sin\theta \quad \cos(\theta+2n\pi)=\cos\theta$

(ii) θ

$\frac{1}{2}\pi$

π

(iii) $\frac{3}{2}\pi$

2π

(iv) $\sin\theta$

$\cos\theta$

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$\sin(-\theta)\equiv-\sin\theta$

$\cos(-\theta)\equiv-\cos\theta$

Q.6. from Euler relation deduce that

(a) $\sin^2\theta+\cos^2\theta\equiv 1\ \left[\text{where } \sin^2\theta\equiv(\sin\theta)^2, \text{etc.}\right]$

$$\sin(\theta_1\pm\theta_2)\equiv\sin\theta_1\cos\theta_2\pm\cos\theta_1\sin\theta_2$$

$$\cos(\theta_1\pm\theta_2)\equiv\cos\theta_1\cos\theta_2\mp\sin\theta_1\sin\theta_2$$

$$\frac{d}{d\theta}\sin\theta=\cos\theta$$

$$\frac{d}{d\theta}\cos\theta=-\sin\theta$$

(b) $\left.\begin{matrix}\phi'(\theta)=\cos\theta\\ \phi''(\theta)=-\sin\theta\\ \phi'''(\theta)=-\cos\theta\\ \phi^4(\theta)=\sin\theta\\ \phi^5(\theta)=\cos\theta\\ \vdots \quad \vdots\end{matrix}\right\}\Rightarrow\left\{\begin{matrix}\phi'(0)=\cos 0=1\\ \phi''(0)=-\sin 0=0\\ \phi'''(0)=-\cos 0=-1\\ \phi^4(0)=\sin 0=0\\ \phi^5(0)=\cos 0=1\\ \vdots \quad \vdots\end{matrix}\right.$

$$\sin\theta=0+\theta+0-\frac{\theta^3}{3!}+0+\frac{\theta^5}{5!}+.....+\frac{\phi^{(n+1)}(p)}{(n+1)!}\theta^{n+1}$$

$$\sin\theta=\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-\frac{\theta^7}{7!}+............(i)$$

$$\psi(\theta)=\cos\theta$$

(c) $\psi(0)=\cos 0=1$

$\left.\begin{matrix}\psi'(\theta)=-\sin\theta\\ \psi''(\theta)=-\cos\theta\\ \psi'''(\theta)=\sin\theta\\ \psi^4(\theta)=\cos\theta\\ \psi^5(\theta)=-\sin\theta\\ \vdots \quad \vdots\end{matrix}\right\}\Rightarrow\left\{\begin{matrix}\psi'(0)=-\sin 0=0\\ \psi''(0)=-\cos 0=-1\\ \psi'''(0)=\sin 0=-1\\ \psi^4(0)=\cos 0=0\\ \psi^5(0)=-\sin 0=1\\ \vdots \quad \vdots\end{matrix}\right.$

$$\cos\theta=1+0-\frac{\theta^2}{2!}+0+\frac{\theta^4}{4!}+.......+\frac{\psi^{(n+1)}(p)}{(n+1)!}\theta^{n+1}$$

Multiplying and dividing by $\cos\theta=1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-\frac{\theta^6}{6!}.......(ii)$

$$e^x=1+x+\frac{1}{2!}x^2+\frac{1}{3!}x^3+\frac{1}{4!}x^4+......$$

$$e^{i\theta}=1+i\theta+\frac{(i\theta)^2}{2!}+\frac{(i\theta)^3}{3!}+\frac{(i\theta)^4}{4!}+\frac{(i\theta)^5}{5!}+........$$

Hence $e^{i\theta}=1+i\theta-\frac{\theta^2}{2!}+\frac{i\theta^3}{3!}+\frac{\theta^4}{4!}+\frac{i\theta^5}{5!}-........$

(d) $e^{i\theta}=\left(1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-.....\right)+i\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-......\right)\rightarrow(a)$

$$e^{-i\theta}=1-i\theta+\frac{(-i\theta)^2}{2!}+\frac{(-i\theta)^3}{3!}+\frac{(-i\theta)^4}{4!}+\frac{(-i\theta)^5}{5!}+....$$

$$e^{-i\theta}=1-i\theta+\frac{\theta^2}{2!}+\frac{i\theta^3}{3!}+\frac{\theta^4}{4!}-\frac{i\theta^5}{5!}-....$$

$$e^{-i\theta}=\left(1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-......\right)-i\left(\theta-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-......\right)\rightarrow(b)$$

$$e^{i\theta}\equiv\cos\theta+i\sin\theta$$

$$e^{-i\theta}\equiv\cos\theta-i\sin\theta$$

θ

Q.7. Find the Cartesian form of each complex number

(a) $\frac{\pi}{6}$

$\frac{\pi}{4}$

$\frac{\pi}{3}$

$\frac{3\pi}{4}$

(b) $\sin\theta$

$\frac{1}{2}$

$$\frac{1}{\sqrt{2}}\left(¿\frac{\sqrt{2}}{2}\right)$$

$$\frac{\sqrt{3}}{2}$$

(c) $\frac{1}{\sqrt{2}}\left(¿\frac{\sqrt{2}}{2}\right)$

$\cos\theta$

$$\frac{\sqrt{3}}{2}$$

$$\frac{1}{\sqrt{2}}\left(¿\frac{\sqrt{2}}{2}\right)$$

Q.8.Find the polar and exponential form

(a) $\frac{1}{2}$

$$\frac{-1}{\sqrt{2}}\left(¿\frac{-\sqrt{2}}{2}\right)$$

$r^2+r+4=0 \qquad a_1=1,\ a_2=4$

$$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-1\pm\sqrt{(-1)^2-4(4)}}{2}$$

$$=\frac{-1\pm\sqrt{-15}}{2}=\frac{-1\pm\sqrt{15}\sqrt{-1}}{2}=\frac{-1}{2}\pm\frac{\sqrt{15}}{2}i$$

$$(\sqrt{-1}=i)$$

$$r_1+r_2=-a_1$$

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$$r_1 r_2 = a_2$$

$$\frac{d\sin\theta}{d\theta} = \cos\theta$$

$$\frac{d^2}{d\theta^2}\sin\theta = \frac{d}{d\theta}\cos\theta = -\sin\theta$$

$$e^{i\pi} = \cos\pi + i\sin\pi$$

$$\cos\pi = -1 \quad and \quad \sin\pi = 0$$

(b) $e^{i\pi} = -1 + i \times 0$

$$e^{i\pi} = -1$$

$$e^{-i\pi/2} = -i$$

$$\theta = \frac{\pi}{2}$$

$$e^{-i\pi/2} = \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} = 0 - i(1) = -i$$

$$5e^{3i\pi/2}$$

$$R = 5 \quad and \quad \theta = \frac{3\pi}{2}$$

$$h = R\cos\theta \quad and \quad v = R\sin\theta$$

$$h = 5\cos\frac{3\pi}{2} \qquad v = 5\sin\frac{3\pi}{2}$$

$$h = 5(0) = 0 \qquad v = 5(-1) = -5$$

$$\cos 3\frac{\pi}{2} = 0$$

$$\sin 3\frac{\pi}{2} = -1$$

$$h + vi = -5i \quad as \ (h = 0)$$

Exercise 15.3:

Q.1.Find the y_p and y_c the general solution and the definite solution of each of the following:

(1) $(1 + \sqrt{3}\ i)$

$$h = 1, \quad v = \sqrt{3}$$

$$R = \sqrt{h^2 + v^2} = \sqrt{(1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

Hence

$$h = R\cos\theta \qquad v = R\sin\theta$$

$$\frac{h}{R} = \cos\theta \qquad \sin\theta = \frac{v}{R}$$

$$\theta = \cos^{-1}\left(\frac{h}{R}\right) \qquad \theta = \sin^{-1}\left(\frac{v}{R}\right)$$

Solving for y_c

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \qquad \theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

Substituting in the above equation

$$R(\cos\theta + i\sin\theta) = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$$

$$Re^{i\theta} = 2e^{i\pi/3}$$

$$r^2 - 3r + 9 = 0$$

$$a = 1, \quad b = -3, \quad c = 9$$

$$r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r_1, r_2 = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(9)}}{2(1)}$$

$$r_1, r_2 = \frac{3 \pm \sqrt{9 - 36}}{2}$$

$$r_1, r_2 = \frac{3 \pm \sqrt{-27}}{2} = \frac{3 \pm \sqrt{27}\sqrt{-1}}{2}$$

$$r_1, r_2 = \frac{3}{2} \pm \frac{3\sqrt{3}}{2} i$$

$$r^2 + 2r + 17 = 0$$

Finding the general solution

$$a = 1, \quad b = 2, \quad c = 17$$

$$r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(17)}}{2(1)}$$

Finding the definite solution

$$r_1, r_2 = \frac{-2 \pm \sqrt{4 - 68}}{2}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{-64}}{2}$$

$$r_1, r_2 = \frac{-2 \pm \sqrt{64}\sqrt{-1}}{2}$$

$$r_1, r_2 = \frac{-2}{2} \pm \frac{8}{2} i$$

$$r_1, r_2 = -1 \pm \frac{8}{2} i = -1 \pm 4i$$

$$2x^2 + x + 8 = 0$$

$$a = 2, \quad b = 1, \quad c = 8$$

$$r_1, r_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hence

$$r_1, r_2 = \frac{-1 \pm \sqrt{(1)^2 - 4(2)(8)}}{2(2)}$$

Verification:

$r_1,\ r_2=\frac{-1\pm\sqrt{1-64}}{4}$

$$r_1,\ r_2=\frac{-1\pm\sqrt{-63}}{4}$$

$r_1,\ r_2=\frac{-1\pm\sqrt{63}\ i}{4}$

Substituting into above

$r_1,\ r_2=\frac{-1\pm 3\sqrt{7}\ i}{4}$

$r_1,\ r_2=\frac{-1}{4}\pm\frac{3\sqrt{7}}{4}i$

$$2x^2-x+1=0$$

$a=2,\quad b=-1,\quad c=1$

(2) $r_1,\ r_2=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$r_1,r_2=\frac{-(-1)\pm\sqrt{(-1)^2-4(2)(1)}}{2(2)}$

Solving for y_p

Let $r_1,r_2=\frac{1}{4}\pm\frac{\sqrt{-7}}{4}=\frac{1}{4}\pm\frac{\sqrt{7}\,i}{4}$

$r_1,\ r_2=\frac{1}{4}\pm\frac{\sqrt{7}}{4}i$

$\pi\ rad=180^0$

$rad=\frac{180^0}{\pi}$

$$\pi\ rad=180^0$$

$\frac{\pi\ rad}{180}=1^0$

$1^0=\frac{\pi\ rad}{180}$

Solving for y_c

Let a'

$\sin^2\theta+\cos^2\theta=1$

Substituting in the above equation

$R^2=h^2+v^2\qquad h=R\ \cos\theta$

$v=R\ \sin\theta$

$R^2=(R\ \cos\theta)^2+(R\ \sin\theta)^2$

$$R^2=R^2\ \cos^2\theta+R^2\ \sin^2\theta$$

$$R^2=R^2\left(\cos^2\theta+\sin^2\theta\right)$$

$\frac{R^2}{R^2}=\cos^2\theta+\sin^2\theta$

$1=\cos^2\theta+\sin^2\theta$

$\sin\frac{\pi}{4}=\cos\frac{\pi}{4}=\frac{1}{\sqrt{2}}$

$\sin(\pi-\frac{3}{4}\pi)=\sin\pi\ \cos\frac{3\pi}{4}-\cos\pi\ \sin\frac{3\pi}{4}$

$\sin(\pi-\frac{3}{4}\pi)=0-(-1)\frac{1}{\sqrt{2}}$

Finding the general solution

$\sin(\pi-\frac{3}{4}\pi)=\frac{1}{\sqrt{2}}$

$ylt=e^{-2t}(A_5\cos 2t+A_6\sin 2t)+\frac{1}{4}$

Finding the definite solution

$y_{(t)}=e^{-2t}(A_5\cos 2t+A_6\sin 2t)+\frac{1}{4}$

Taking derivative W.r.t‘t’

$y'_{(t)}=e^{-2t}(-2A_5\sin 2t+2A_6\cos 2t)$

$y'_{(t)}=-2e^{-2t}(A_5\cos 2t+2A_6\sin 2t)$

$y_{(0)}=A_5+\frac{1}{4}=\frac{9}{4}$

$y'_{(0)}=2A_6-2A_5=4$

$A_5=\frac{9}{4}-\frac{1}{4}=\frac{8}{4}=2$

$A_5=2$

$y'_{(0)}=2A_6-2(2)=4$

$A_6=\frac{4+4}{2}=\frac{8}{2}=4$

$y_{(t)}=e^{-2t}(2\cos 2t+4\sin 2t)+\frac{1}{4}$

(3) $y''_{(t)}+3y'_{(t)}+4y=12$

$y_{(0)}=2$

$y'_{(0)}=2$

Solving for y_p

Let $y_{(t)}=k$

Then $y'_{(t)}=y''_{(t)}=0$

$4k=12$

$k=\frac{12}{4}=3$

$y_p=3$

Solving for y_c

Let $y_{(t)}=Ae^{rt}$

$y'_{(t)}=rAe^{rt}$

$y''_{(t)}=r^2 Ae^{rt}$

$r^2 Ae^{rt}+3rAe^{rt}+4Ae^{rt}=0$

$Ae^{rt}(r^2+3r+4)=0$

$r^2+3r+4=0$

$$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}$$

$$r_1,\ r_2=\frac{-3\pm\sqrt{(3)^2-4(4)}}{2}$$

$$r_1,\ r_2=\frac{-3\pm\sqrt{-7}}{2}=\frac{-3\pm\sqrt{7}i}{2}$$

$$r_1,\ r_2=\frac{-3}{2}\pm\frac{\sqrt{7}}{2}i$$

$h=-\frac{3}{2}$ $\qquad v=\frac{\sqrt{7}}{2}$

$$y_c=e^{-3/2t}(A_5\cos\frac{\sqrt{7}}{2}t+A_6\sin\frac{\sqrt{7}}{2}t$$

Finding the general solution

$y_t=y_c+y_p$

$$=e^{-3/2t}(A_5\cos\frac{\sqrt{7}}{2}t+A_6\sin\frac{\sqrt{7}}{2}t)+3$$

Finding the definite solution

$$y_{(t)}=e^{-3/2t}(A_5\cos\frac{\sqrt{7}}{2}t+A_6\sin\frac{\sqrt{7}}{2}t)$$

$$y'_{(t)}=e^{-3/2t}(-\frac{\sqrt{7}}{2}A_5\sin\frac{\sqrt{7}}{2}t+\frac{\sqrt{7}}{2}A_6\cos\frac{\sqrt{7}}{2}t)$$

$$y'_{(t)}=\frac{-3}{2}e^{-3/2t}(A_5\cos\frac{\sqrt{7}}{2}t+A_6\sin\frac{\sqrt{7}}{2}t)$$

$y_{(0)}=A_5+3=2$

$y_{(0)}=A_5=2-3=-1$

$$y'_{(0)}=\frac{\sqrt{7}}{2}A_6-\frac{3}{2}A_5=2$$

$$\frac{\sqrt{7}}{2}A_6-\frac{3}{2}(-1)=2$$

$$A_6=(2-\frac{3}{2})\frac{2}{\sqrt{7}}$$

$$A_6=\frac{1}{2}\ \frac{2}{\sqrt{7}}$$

$$A_6=\frac{1}{\sqrt{7}}$$

$$a_6=\frac{\sqrt{7}}{7}$$

Note

$$\frac{1}{\sqrt{7}}\cdot\frac{\sqrt{7}}{\sqrt{7}}=\frac{\sqrt{7}}{7}$$

see table b

$$y_{(t)}=e^{-3/2t}(-\cos\frac{\sqrt{7}}{2}t+\frac{\sqrt{7}}{7}\sin\frac{\sqrt{7}}{2}t)+3$$

Q.4 $y''_{(t)}-2y'_{(t)}+10y=5$

Let $y_t=k$

Then $y'_{(t)}=y''_{(t)}=0$

$10k=5$

$y_p=k=5/10$

$y_p=1/2$

Finding y_c

Let $y_{(t)}=Ae^{rt}$

$y'_{(t)}=rAe^{rt}$

$y''_{(t)}=r^2 Ae^{rt}$

$r^2 Ae^{rt}-2rAe^{rt}+10Ae^{rt}=0$

$Ae^{rt}(r^2-2r+10)=0$

$r^2-2r+10=0$

$$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}$$

$$r_1,\ r_2=\frac{2\pm\sqrt{(-2)^2-4(10)}}{2}$$

$$r_1,\ r_2=\frac{2\pm\sqrt{4-40}}{2}=\frac{2\pm\sqrt{36}i}{2}$$

$$r_1,\ r_2=\frac{2}{2}\pm\frac{6}{2}i=1\pm3i$$

$h=1$ $\qquad v=3$

$y_c=e^{ht}(A_5\cos vt+A_6\sin vt)$

$y_c=e^{t}(A_5\cos 3t+A_6\sin 3t)$

Finding the general solution

$y_{(t)}=y_c+y_p$

$$=e^{t}(A_5\cos 3t+A_6\sin 3t)+\frac{1}{2}$$

Finding the definite solution

$$y_{(t)}=e^{t}(A_5\cos 3t+A_6\sin 3t)+\frac{1}{2}$$

$$y'_{(t)}=e^{t}(-3A_5\sin 3t+3A_6\cos 3t)+e^{t}(A_5\cos 3t+A_6\sin 3t)$$

$$y_{(0)}=A_5+\frac{1}{2}=6 \qquad A_5=5\frac{1}{2}=\frac{11}{2}$$

$$y'_{(0)}=3A_6+A_5=\frac{17}{2}$$

$$3A_6=\frac{17}{2}-\frac{11}{2}=\frac{6}{2}=3$$

$$A_6=\frac{3}{3}=1$$

$$y_{(t)}=e^{t}\left(5\frac{1}{2}\cos 3t+\sin 3t\right)+\frac{1}{2} \rightarrow \textit{definite solution}$$

Q.5 $y''_{(t)}+9y'_{(t)}=3 \qquad y'_{(0)}=1 \quad y'_{(0)}=3$

For y_p

Let $y=k$ then $y'_{(t)}=y''_{(t)}=0$

$$9k=3$$

$$k=\frac{3}{9}=\frac{1}{3}$$

$$y_p=\frac{1}{3}$$

For y_c

Let $y_{(t)}=Ae^{rt}$

$$y'_{(t)}=rAe^{rt}$$

$$y''_{(t)}=r^2Ae^{rt}$$

Substituting into reduced equation

$$r^2Ae^{rt}+9Ae^{rt}=0$$

$$Ae^{rt}(r^2+9)=0$$

$$r^2+9=0$$

$$a_1=0 \qquad a_2=9$$

$$r_1,\, r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}$$

$$r_1,\, r_2=\frac{0\pm\sqrt{-4(a)}}{2}=0\pm\frac{6i}{2}=0\pm 3i$$

$$h=0 \qquad v=3$$

$$y_c=e^{ht}(A_5\cos vt+A_6\sin vt)$$

$$y_c=e^{0}(A_5\cos 3t+A_6\sin 3t)$$

$$y_c=(A_5\cos 3t+A_6\sin 3t)$$

$$y_{(t)}=y_c+y_p$$

$$y_{(t)}=(A_5\cos 3t+A_6\sin 3t)+\frac{1}{3} \quad \textit{General solution}$$

Finding definite solution

$$y_{(t)}=(A_5\cos 3t+A_6\sin 3t)+\frac{1}{3}$$

$$y'_{(t)}=(-3A_5\sin 3t+3A_6\cos 3t)$$

$$y_{(0)}=A_5+\frac{1}{3}=1 \qquad A_5=\frac{2}{3}$$

$$y'_{(0)}=3A_6=3$$

$$A_6=3/3=1 \Rightarrow A_6=1$$

$$y_{(t)}=\left(\frac{2}{3}\cos 3t+\sin 3t\right)+\frac{1}{3}$$

Q.6 $2y''_{(t)}-12\{y'_{(t)}+20y=40$ ¿

$$y_{(0)}=4$$

$$y'_{(0)}=5$$

$$y''_{(t)}-6y'_{(t)}+10y=20$$

For y_p

Let $y_{(t)}=k$

$$y'_{(t)}=y''_{(t)}=0$$

$$10k=20 \Rightarrow k=20/10=2$$

$$y_p=2$$

For y_c

Let $y_{(t)}=Ae^{rt}$

Then $y'_{(t)}=rAe^{rt}$ and $y''_{(t)}=r^2Ae^{rt}$

Substituting into deduced equation

$$y''_{(t)}-6y'_{(t)}+10y=0$$

$$r^2Ae^{rt}-6rAe^{rt}+10Ae^{rt}=0$$

$$Ae^{rt}(r^2-6r+10)=0$$

$$r^2-6r+10=0$$

$$a_1=-6 \qquad a_2=10$$

$$r_1,\, r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}$$

$$r_1,\, r_2=\frac{6\pm\sqrt{(-6)^2-4(10)}}{2}=\frac{6\pm\sqrt{36-40}}{2}=\frac{6\pm\sqrt{-4}}{2}$$

$$r_1,\, r_2=\frac{6\pm 2i}{2}=3\pm i$$

$$h=3 \qquad v=1$$

$$y_c=e^{ht}(A_5\cos vt+A_6\sin vt)$$

$$y_c=e^{3t}(A_5\cos t+A_6\sin t)$$

$y_{(t)} = y_c + y_p$

$y_{(t)} = e^{3t}(A_5 \cos t + A_6 \sin t) + 2$

$y'_{(t)} = e^{3t}(-A_5 \sin t + A_6 \cos t) + 3e^{3t}(A_5 \cos t + \sin t)$

$y_{(0)} = A_5 + 2 = 4 \qquad A_5 = 2$

$y'_{(0)} = A_6 + 3A_5 = 5$

$A_6 = 5 - 3(2) = -1$

$y_t = e^{3t}(2\cos t - \sin t) + 2$

Which of the above six differential equation yield time paths with?

(a) Demand fluctuations
(b) Uniform fluctuations
(c) Explosive fluctuations

(1) $As \;\; t \to \infty \quad e^{2t} \to \infty$ And time path shows explosive fluctuations

(2) $As \;\; t \to \infty \quad e^{-2t} \to 0$ And time path will shows damped fluctuations

(3) $As \;\; t \to \infty \quad e^{-3/2\,t} \to 0$ And time path will shows damped fluctuations

(4) $As \;\; t \to \infty \quad e^{t} \to \infty$ And time path will shows explosive fluctuations

(5) $As \;\; h = 0 \quad hence \quad e^{ht} = 1$ And time path will shows uniform fluctuations

(6) $As \;\; t \to \infty \quad e^{3t} \to \infty$ And time path will shows explosive fluctuations

A Market Model with Price Expectations:

Some ties buyers and sellers may base their market behavior not only on the current price but also on the price tend prevailing at the time. For the price trend is likely to lead then to certain expectations regarding the price level in the future and these expectations can in turn influence their demand and supply decisions.

Price trend and Price Expectations:

In the continuous time context the price trend information is to be found primarily in the two derivatives $\frac{dp}{dt}$ (whether price is rising) and $\frac{d^2p}{dt^2}$ (whether increasing at an increasing rate). To take the price trend into account let us now include and supply functions.

$Q_d = \alpha - \beta p + m p' + n p'' \quad (\alpha,\ \beta > 0)$

$Q_s = -\alpha + 8p + u p' + w p'' \quad (\gamma,\ \delta > 0)$

If m is greater than zero, for instance a rising price will cause Q_d to increase this would suggest that buyers expect the rising price to continue to rise and hence prefer to increase their purchase now when the price is still relatively low. The inclusion of the parameter n makes the buyers behavior depend also on the rate of change of $\frac{dp}{dt}$. The parameter u and w carry a similar implication on the seller's side of the picture.

A Simplified Model:

For simplicity we shall assume that only the demand function contains price expectations. Specifically we let m and n be nonzero. But let u=w=0. Further assume that market is clear at every point of time. Then we may equate the demand and supply functions to obtain the differential equation, i.e.

$Q_d = Q_s$

$\alpha - \beta p + m p' + n p'' = -\gamma + \delta p \qquad u = w = 0$

$n p'' + m p' - \beta p - \delta p = -\gamma - \alpha = -\gamma - \alpha$

$n p'' + m p' - (\beta + \delta) p = -(\alpha + \gamma)$

$p'' + \frac{m}{n} p' - \frac{(\beta+\delta)}{n} p = -\frac{(\alpha+\gamma)}{n}$

Where

$a_1 = \frac{m}{n} \qquad a_2 = \frac{-(\beta+\delta)}{n} \qquad b = \frac{-(\alpha+\delta)}{n}$

The Time Path of Price:

The inter temporal price of the model the particular integral p_p is easily found as follows.

$$p_p=\frac{b}{a_2}=\frac{\alpha+\gamma}{\beta+\delta}$$

Since it is a constant it represents a stationary equilibrium. As for the complementary function p_c there are three possible cases.

Case 1 (distinct real roots):

$$\left(\frac{m}{n}\right)^2>-4\left(\frac{\beta+\delta}{n}\right)$$

This complementary function of this case is

$$p_c=A_1e^{r_1t}+A_2e^{r_2t}$$

$$r_1,\ r_2=\frac{1}{2}\left[-\frac{m}{2}\pm\sqrt{\left(\frac{m}{n}\right)^2+4\left(\frac{\beta+\delta}{n}\right)}\right]\rightarrow(a)$$

According the general solution is

$$p_{(t)}=p_c+p_p=A_1e^{r_1t}+A_2e^{r_2t}+\frac{\alpha+\gamma}{\beta+\delta}\rightarrow(b)$$

Case 2(double real roots):

$$\left(\frac{m}{n}\right)^2=-4\left(\frac{\beta+\delta}{n}\right)$$

In this case, the characteristic root takes the single value

$$r=-\frac{m}{2n}$$

Thus general solution may be written as

$$p_{(t)}=A_3e^{\frac{-mt}{2n}}+A_4te^{\frac{-mt}{2n}}+\frac{\alpha+\gamma}{\beta+\delta}$$

Case 3 (complex roots):

$$\left(\frac{m}{n}\right)^2<-4\left(\frac{\beta+\delta}{n}\right)$$

In this case the characteristic roots are the pair of conjugate complex numbers

$$r_1,\ r_2=h\pm vi$$

Where

$$h=-\frac{m}{2n}\quad and\quad v=\frac{1}{2}\sqrt{-4\left(\frac{\beta+\delta}{n}\right)-\left(\frac{m}{n}\right)^2}$$

And the general solution will be

$$p_{(t)}=e^{\frac{-mt}{2n}}\left(A_5\cos vt+A_6\sin vt\right)+\frac{\alpha+\gamma}{\beta+\delta}$$

General Conclusion:

A couple of general conclusion can be deduced from these results. First, if n>0, $-4\frac{(\beta+\delta}{n}$ must be negative and hence less than $\left(\frac{m}{n}\right)^2$. Hence case 2 and 3 can immediately be ruled out. Moreover with n positive (as are β and δ), the expression under the square root sign in (a) necessarily exceeds $\left(\frac{m}{n}\right)^2$ and thus the square root must be greater than $\left(\frac{m}{n}\right)$. The ± sign in (a) would than produce one positive root (r_1) and one negative root (r_2). Consequently the inter temporal equilibrium is dynamically unstable, unless the definitize value of the constant A_1 happens to be zero in (b) Second if n<0 then all the three cases becomes feasible. Under case 1 we can be sure that both roots will be negative if m is negative. Interestingly the repeated root of case 2 will also be negative if m is negative. Moreover since h the real part of the complex root in case 3, takes the same value as the repeated root r in case 2, the negatively of m will also guarantee that his negative. In short, for all the three cases, the dynamic stability of equilibrium is ensured when the parameters m and n are both negative.

Example 1:

$$Q_d=42-4p-4p'+p''$$

$$Q_s=-6+8p$$

Assuming that market is clear at every point of time then

$Q_d = Q_s$

$42-4p-4p'+p''=-6+8p$

$p''-4p'-8p-4p+42+6=0$

$p''-4p'-12p=-48$

$p_p=\dfrac{-48}{-12}=4$

For complementary solution

$r^2-4r-12=0 \qquad a_1=-4$

$a_2=-12$

$r_1,\ r_2=\dfrac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}$

$r_1,\ r_2=\dfrac{4\pm\sqrt{(-4)^2-4(-12)}}{2}=\dfrac{4\pm\sqrt{16+48}}{2}$

$r_1,\ r_2=\dfrac{4\pm 8}{2}=6,\ -2$

$r_1=6 \qquad r_2=-2$

$p_c=A_1e^{r_1t}+A_2e^{r_2t}$

$p_c=A_1e^{6t}+A_2e^{-2t}$

$p_{(t)}=p_c+p_p=A_1e^{6t}+A_2e^{-2t}+4$

General solution

$p'_{(t)}=6A_1e^{6t}+A_2e^{-2t}$

$p_{(0)}=A_1+A_2+4=6 \quad \rightarrow (i)$

$p'_{(0)}=6A_1-2A_2=4 \quad \rightarrow (ii)$

Multiplying by 2 equation in 1

$2A_1+2A_2=4$

$6A_1-2A_2=4$

$8A_1 \quad =8$

$A_1=\dfrac{8}{8}=1$

From equation (ii)

$6(1)-2(A_2)=4$

$-2A_2=4-6=-2$

$A_2=\dfrac{-2}{2}=+1$

$p_t=e^{6t}+e^{-2t}+4$

As $t\rightarrow\infty \quad e^{6t}\rightarrow\infty$ hence inter temporal equilibrium is dynamically unstable.

Example 2:

$Q_d=40-2p-2p'-p''$

$Q_s=-5+3p$

$Q_d=Q_s$

$40-2p-2p'-p''=-5+3p$

$-p''-2p'-2p-3p=-5-40$

$p''+2p'+5p=45$

$p_p=\dfrac{45}{5}=9$

For p_c

$r^2+2r+5=0$

$r_1,\ r_2=\dfrac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\dfrac{-2\pm\sqrt{(2)^2-4(5)}}{2}$

$r_1,\ r_2=\dfrac{-2\pm\sqrt{4-20}}{2}=\dfrac{-2\pm 4i}{2}=-1\pm 2i$

$h=-1 \qquad v=2$

$p_c=e^{ht}(A_5\cos vt+A_6\sin vt)$

$p_c=e^{-t}(A_5\cos 2t+A_6\sin 2t)$

$p_t=p_c+p_p$

$p_t=e^{-t}(A_5\cos 2t+A_6\sin 2t)+9 \rightarrow$ *General solution*

$p'_t=e^{-t}(-2A_5\sin 2t+2A_6\cos 2t)$

$\qquad -e^{-t}(A_5\cos 2t+A_6\sin 2t)$

$p_{(0)}=A_5+9=12 \qquad A_5=12-9=3$

$p'_{(0)}=2A_6A_5=1 \qquad A_6=2$

$p_{(t)}=e^{-t}(3\cos 2t+2\sin 2t)+9$

As $t\rightarrow\infty \quad e^{-t}=0$ Hence inter temporal is dynamically stable.

Exercise 15.4:

Q.1. let the parameter m, n v and w all be non zero

(a) Assuming market clearance at every point of time, write the new differential equation of the model.

(b) Find the inter temporal equilibrium price

(c) Under what circumstance can periodic fluctuation are ruled out.

(a) $Q_d=\alpha-\beta p+mp'+np'' \qquad (\alpha,\ \beta>0)$

$Q_s=-\alpha+\delta p+up'+wp'' \qquad (\gamma,\ \delta>0)$

$Q_d=Q_s$

$\alpha-\beta p+mp'+np''=-\gamma+\delta p+up'+wp''$

$$np'' - wp'' + mp' - up' - \beta p - \delta p = -\alpha - \gamma$$
$$(n-w)p'' + (m-u)p' - (\beta-\delta)p = -(\alpha+\gamma)$$
$$p'' + \frac{(m-u)}{(n-w)}p' - \frac{(\beta+\delta)}{(n-w)}p = -\frac{(\alpha+\gamma)}{(n-w)}$$

(b) $$p_p = \frac{\frac{-(\alpha+\gamma)}{n-w}}{\frac{-(\beta+\delta)}{n-w}} = \frac{b}{a_2}$$

$$p_p = \frac{\alpha+\gamma}{\beta+\delta}$$

(c) The periodic fluctuation can be ruled out in following conditions

$$a_1^2 \ge 4a_2$$
$$\left(\frac{m-u}{n-w}\right)^2 \ge -4\left(\frac{(\beta+\delta)}{(n-w)}\right)$$
$$or \quad n-w>0$$
$$or \quad n>w$$

Q.2. Let the demand and supply function are as follows

$$Q_d = \alpha - \beta p + mp' + np''$$
$$Q_s = -\gamma + \delta p$$

(a) If the markets is not always cleared but adjust according to

$$\frac{dp}{dt} = j(Q_d - q_s) \quad (j>0)$$

Write the appropriate new differential equation

$$p'_{(t)} = j(Q_d - Q_s)$$
$$p'_{(t)} = j(\alpha - \beta p + mp' + np'' + \gamma - \delta p)$$
$$p' = j\alpha - j\beta p + jm\{p' + jn\{p¿'' + j\gamma - j\delta p)¿$$
$$jn\{p'' + jm\{p¿' - p' - j\beta p - j\delta p = -j(\alpha+\gamma)¿$$
$$jn\{p'' + (jm-1)p' - j(\beta+\delta)p = -j(\alpha+\gamma)¿$$
$$p'' + \frac{(jm-1)}{jn}p' - \frac{j(\beta+\delta)}{jn}p = \frac{-j(\alpha+\gamma)}{jn}$$
$$p'' + \frac{(jm-1)}{jn} - \frac{(\beta+\delta)}{n}p = -\frac{(\alpha+\gamma)}{n}$$

(b) Find the inter temporal equilibrium price and market clearing equilibrium price.

(i) $p_p = \dfrac{\alpha+\gamma}{\beta+\delta}$ Inter temporal equilibrium price.

(ii) Market clearing equilibrium price

$$Q_d = Q_s$$
$$\alpha - \beta p + mp' + np'' = -\gamma + \delta p$$
$$np'' + mp' - \beta p - \delta p = -\alpha - \gamma$$
$$np'' + mp' - (\beta+\delta)p = -(\alpha+\gamma)$$
$$-(\beta+\delta)p = -(\alpha+\gamma) - mp' - np''$$
$$\bar{p} = \frac{(\alpha+\gamma) - mp' - np''}{\beta+\delta}$$

(c) State the condition for having a fluctuating price path. Can fluctuation occurs if n>o?

The conditions for having a fluctuating price path will be

$$a_1^2 < 4a_2$$
$$\left(\frac{jm-1}{jn}\right)^2 < -4\left(\frac{\beta+\delta}{n}\right)$$

If n>0 no periodic fluctuations

If n<0 fluctuations will occur

Q.3. let the demand and supply be

$$Q_d = 9 - p + p' + 3p'' \qquad p_{(0)} = 4$$
$$Q_s = -1 + 4p - p' + 5p'' \qquad p'_{(0)} = 4$$

(a) Find the price path assuming market clearance at every point of time

$$Q_d = Q_s$$
$$9 - p + p' + 3p'' = -1 + 4p - p' + 5p''$$
$$5p'' - 3p'' - p' - p' + 4p + p = 1 + 9$$
$$2p'' - 2p' + 5p = 10$$
$$p'' - p' + \frac{5}{2}p = 5 \qquad a_2 = \frac{5}{2}, \quad a_1 = -1$$
$$b = 5$$
$$p_p = \frac{b}{a_2} = \frac{5}{5/2} = 2$$

For p_c

$$r^2 - r + \frac{5}{2} = 0$$
$$r_1, r_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2} = \frac{1 \pm \sqrt{(-1)^2 - 4(\frac{5}{2})}}{2}$$
$$r_1, r_2 = \frac{1 \pm \sqrt{1-10}}{2} = \frac{1 \pm 3i}{2} = \frac{1}{2} \pm \frac{3}{2}i$$
$$h = \frac{1}{2} \qquad v = \frac{3}{2}$$

$$p_c = e^{ht}\left(A_5 \cos vt + A_6 \sin vt\right)$$

$$p_c = e^{\frac{1}{2}t}\left(A_5 \cos \frac{3}{2}t + A_6 \sin \frac{3}{2}t\right)$$

$$p_t = p_c + p_p$$

$$p_t = e^{\frac{1}{2}t}\left(A_5 \cos \frac{3}{2}t + A_6 \sin \frac{3}{2}t\right) + 2$$

$$p_t^{'} = e^{\frac{1}{2}t}\left(-\frac{3}{2}A_5 \sin \frac{3}{2}t + \frac{3}{2}A_6 \cos \frac{3}{2}t\right) + \frac{1}{2}e^{t/2}$$

$$\left(A_5 \cos \frac{3}{2}t + A_6 \sin \frac{3}{2}t\right)$$

$$p_{(0)} = A_5 + 2 = 4 \qquad A_5 = 2$$

$$p_{(0)}^{'} = \frac{3}{2}A_6 + \frac{1}{2}A_5 = 4$$

$$p_{(0)}^{'} = \frac{3}{2}A_6 + \frac{1}{2}(2) = 4$$

$$A_6 = \frac{3}{3}(2) = 2$$

Hence $A_5 = 2, \quad A_6 = 2$

$$p_{(t)} = e^{\frac{1}{2}t}\left(2\cos \frac{3}{2}t + 2\sin \frac{3}{2}t\right) + 2$$

(b) Is the time path convergent with fluctuation?

Answer:

As $t \to \infty \quad e^{t/2} \to \infty$ and time path will be divergent with explosive fluctuation

Example 1:

Let the three equations of the model take the specific form

$$p = \frac{1}{6} - 3u + \pi \qquad \to (i)$$

$$\frac{d\pi}{dt} = \frac{3}{4}(p - \pi) \qquad \to (ii)$$

$$\frac{du}{dt} = -\frac{1}{2}(m - p) \qquad \to (iii)$$

Find the time path of π, p and u starting from equation (ii)

$$\frac{d\pi}{dt} = \frac{3}{4}(p - \pi)$$

Substituting (i) in (ii)

$$\frac{d\pi}{dt} = \frac{3}{4}\left[\frac{1}{6} - 3u + \pi\right] - \frac{3}{4}\pi$$

$$\frac{d\pi}{dt} = \frac{1}{8} - \frac{9}{4}u + \frac{3}{4}\pi - \frac{3}{4}\pi$$

$$\frac{d}{dt}\left(\frac{d\pi}{dt}\right) = \frac{d}{dt}\left(\frac{1}{8} - \frac{9}{4}u\right)$$

$$\frac{d^2\pi}{dt^2} = -\frac{9}{4}\frac{du}{dt} \qquad \to (iv)$$

Substituting (iii) in (iv)

$$\frac{d^2\pi}{dt^2} = -\frac{9}{4}\left[-\frac{1}{2}(m - p)\right]$$

$$\frac{d^2\pi}{dt^2} = -\frac{9}{4}\left[-\frac{1}{2}m + \frac{1}{2}p\right]$$

$$\frac{d^2\pi}{dt^2} = \frac{9}{8}m - \frac{9}{8}p \qquad \to (v)$$

Solving equation (ii) for p

$$\frac{d\pi}{dt} = \frac{3}{4}p - \frac{3}{4}\pi \quad \Rightarrow \frac{3}{4}p = \frac{d\pi}{dt} + \frac{3}{4}\pi$$

$$p = \frac{4}{3}\frac{d\pi}{dt} + \pi \qquad \to (vi)$$

Substituting (vi) in (v)

$$\frac{d^2\pi}{dt^2} = \frac{9}{8}m - \frac{9}{8}\left[\frac{4}{3}\frac{d\pi}{dt} + \pi\right]$$

$$\frac{d^2\pi}{dt^2} = \frac{9}{8}m - \frac{3}{2}\frac{d\pi}{dt} - \frac{9}{8}\pi$$

$$\frac{d^2\pi}{dt^2} + \frac{3}{2}\frac{d\pi}{dt} + \frac{9}{8}\pi = \frac{9}{8}m$$

This is the 2nd order differential equation in variable π. finding particular integral and complementary function

$$\pi_p = \frac{b}{a_2} = \frac{9/8\,m}{9/8} = m$$

For π_c using characteristic equation

$$r^2 + \frac{3}{2}r + \frac{9}{8} = 0 \qquad a_1 = \frac{3}{2}; \quad a_2 = \frac{9}{8}$$

$$r_1,\ r_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2} = \frac{1}{2}\left[-\frac{3}{2} \pm \sqrt{\left(\frac{3}{2}\right)^2 - 4\left(\frac{9}{8}\right)}\right]$$

$$r_1,\ r_2 = \frac{1}{2}\left[-\frac{3}{2} \pm \sqrt{\frac{9}{4} - \frac{9}{2}}\right] = \frac{1}{2}\left[-\frac{3}{2} \pm \sqrt{\frac{9 - 18}{4}}\right]$$

$$r_1,\ r_2 = \frac{1}{2}\left[-\frac{3}{2} \pm \sqrt{\frac{9}{4}}\, i\right] = \frac{1}{2}\left[-\frac{3}{2} \pm \frac{3}{2}i\right]$$

$$r_1,\ r_2=-\frac{3}{4}\pm\frac{3}{4}i$$

$$h=-\frac{3}{4} \qquad v=\frac{3}{4}$$

$$\pi_c=e^{ht}(A_5\cos vt+A_6\sin vt)$$

$$\pi_c=e^{-\frac{3}{4}t}(A_5\cos\frac{3}{4}t+A_6\sin\frac{3}{4}t)$$

$$\pi_t=\pi_c+\pi_p$$

$$\pi_t=e^{\frac{-3}{4}t}(A_5\cos\frac{3}{4}t+A_6\frac{3}{4}t)+m$$

As $t\to\infty \quad \pi_c\to 0 \quad \pi_t\to m$ so solution is dynamically stable and time path is convergent with damped fluctuations

Finding the time path of p by using equation (vi)

$$p=\frac{4}{3}\frac{d\pi}{dt}+\pi$$

This equation requires $\frac{d\pi}{dt}$ and π

$$\frac{d\pi}{dt}=e^{\frac{-3}{4}t}(\frac{3}{4}A_6\cos\frac{3}{4}t-\frac{3}{4}A_5\sin\frac{3}{4}t)-\frac{3}{4}e^{\frac{-3}{4}t}(A_5\cos\frac{3}{4}t+A_6\sin\frac{3}{4}t)$$

$$p_{(t)}=e^{\frac{-3}{4}t}(A_6\cos\frac{3}{4}t-A_5\sin\frac{3}{4}t)-e^{\frac{-3}{4}t}(A_5\cos\frac{3}{4}t+A_6\sin\frac{3}{4}t)+e^{\frac{-3}{4}t}(A_5\cos\frac{3}{4}t+A_6\sin\frac{3}{4}t)+m$$

$$p_{(t)}=e^{\frac{-3}{4}t}(A_6\cos\frac{3}{4}t-A_5\sin\frac{3}{4}t)+m$$

As $t\to\infty \quad p_c\to 0 \quad and \quad p_t\to m$ this solution is dynamically stable and time path is convergent with damped fluctuations. Finding the time path of u by using (i)

$$p=\frac{1}{6}-3u+\pi$$

$$3u=\frac{1}{6}-p+\pi$$

$$u=\frac{1}{18}-\frac{1}{3}p+\frac{1}{3}\pi$$

$$u=\frac{1}{18}-\frac{1}{3}e^{\frac{-3}{4}t}(A_6\cos\frac{3}{4}t-A_5\sin\frac{3}{4}t)-\frac{1}{3}m+\frac{1}{3}e^{\frac{-3}{4}t}(A_5\cos\frac{3}{4}t+A_6\sin\frac{3}{4}t)+\frac{1}{3}m$$

$$u_{(t)}=\frac{1}{3}e^{-3t/4}(B_1\cos\frac{3}{4}t+B_2\sin\frac{3}{4}t)+\frac{1}{18}$$

$$\begin{vmatrix}B_1=A_5-A_6\\ B_2=A_5+A_6\end{vmatrix}$$

As $t\to\infty \quad u_{(t)}\to\frac{1}{18}$ and equilibrium is dynamically stable time path is convergent with damped fluctuation

Exercise 15.5:

Q.1. In the inflation unemployment model retain (15.33) and (15.34) but delete (15.35) and let u be exogenous instead.

(a) What kind of differential equation will now exist?

$$p=\alpha-T-\beta u+h\pi \qquad \to(i)$$

$$\frac{d\pi}{dt}=j(p-\pi) \qquad \to(ii)$$

Put (i) into (ii)

$$\frac{d\pi}{dt}=j(\alpha-T-\beta u+h\pi-\pi)$$

$$\frac{d\pi}{dt}=j\alpha-jT-j\beta u+jh\pi-j\pi$$

$$\frac{d\pi}{dt}=j(\alpha-T-\beta u)-j(1-h)\pi$$

$$\frac{d\pi}{dt}+j(1-h)\pi=j(\alpha-T-\beta u)$$

It is the first order differential equation

(b) How many differential equations can you obtain?

It is possible now to have periodic fluctuation in the complementary function?

Answer:

In first order differential equation, we can obtain only one characteristic root and no periodic fluctuations

Q.2. In the text discussion, we condensed the inflation unemployment model into a differential equation in the variable π. Show that model can be condensed into a second order differential equation in variable u, with the same a_1 an a_2 co-efficient as in (15.37 {)''¿but a differential constant term $b = kj(\alpha - T - (1-h)m$

Model:

$$p = \alpha - \beta u - T + h\pi \quad \rightarrow (i)$$

$$\frac{d\pi}{dt} = j(p - \pi) \quad \rightarrow (ii)$$

$$\frac{du}{dt} = -k(m - p) \quad \rightarrow (iii)$$

Substituting (i) in (iii)

$$\frac{du}{dt} = -km + k(\alpha - \beta u - T + h\pi)$$

Taking derivative to create place for $\frac{d\pi}{dt}$

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + kh\frac{d\pi}{dt}$$

Substituting the value of $\frac{d\pi}{dt}$ from equation (ii)

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + kh[j(p - \pi)]$$

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + khj(p - \pi) \quad \rightarrow (iv)$$

Solving equation (iii) for p

$$\frac{du}{dt} = -k(m - p)$$

$$\frac{du}{dt} = -km + kp$$

$$p = m + \frac{1}{k}\frac{du}{dt} \quad \rightarrow (v)$$

Taking equation (i) and substituting (v) in (i)

$$p = \alpha - \beta u - T + h\pi$$

$$m + \frac{1}{k}\frac{du}{dt} = \alpha - \beta u - T + h\pi$$

$$h\pi = \frac{1}{k}\frac{du}{dt} + \beta u + m + T - \alpha$$

$$\pi = \frac{1}{kh}\frac{du}{dt} + \beta u + \frac{m}{h} + \frac{T}{h} - \frac{\alpha}{h} \quad \rightarrow (vi)$$

Putting equation (v) and (vi) in equation (iv)

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + khj(p - \pi)$$

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + khj \left[m + \frac{1}{k}\frac{du}{dt} - \frac{1}{h}\left(\frac{1}{k}\frac{du}{dt} + \beta u + m + T - \alpha\right)\right]$$

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + mkhj + hj\frac{du}{dt} - j\frac{du}{dt} - kj\beta u - kjm - kjT + kj\alpha$$

$$\frac{d^2u}{dt^2} = -k\beta\frac{du}{dt} + hj\frac{du}{dt} - j\frac{du}{dt} - kj\beta u + mkhj - kjm - kjT + kj\alpha$$

$$\frac{d^2u}{dt^2} = -[\beta k + j - hj]\frac{du}{dt} - kj\beta u + kj(\alpha - T - m + mh)$$

$$\frac{d^2u}{dt^2} = -[\beta k + j(1-h)]\frac{du}{dt} - kj\beta u + kj[\alpha - T - m(1-h)]$$

$$\frac{d^2u}{dt^2} = -[\beta k + j(1-h)]\frac{du}{dt} + kj\beta u = kj[\alpha - T - m(1-h)]$$

Q.3. Let the adaptive expectation hypothesis is replaced by so called perfect foresight hypothesis π = p but retain (15.33) and (15.35)

(a) Derive a differential equation in variable p.

$$\pi = p$$

$$p = \alpha - \beta u - T + h\pi \quad \rightarrow (i)$$

$$\frac{du}{dt} = -k(m - p) \quad \rightarrow (ii)$$

When p = π then equation (i) becomes

$$p = \alpha - \beta u - T + hp$$

Taking derivative W.r.t. "t" to create place for $\frac{du}{dt}$

$$\frac{dp}{dt} = -\beta\frac{du}{dt} + h\frac{dp}{dt}$$

$$(1-h)\frac{dp}{dt} = -\beta\frac{du}{dt}$$

Substituting from equation (ii)

$(1-h)\frac{dp}{dt}=-\beta[-k(m-p)]$

$(1-h)\frac{dp}{dt}=\beta km-\beta kp$

$\frac{dp}{dt}=\frac{\beta km}{(1-h)}-\frac{\beta k}{(1-h)}p$

$\frac{dp}{dt}+\frac{\beta k}{(1-h)}p=\frac{\beta km}{(1-h)} \qquad (Here \quad h\neq 1)$

This is differential equation (first order) in variable p.

(b) Derive differential equation in the variable u.

When π = p equation (i) becomes

$p=\alpha-\beta u-T+hp$

Put value of p in equation (ii)

$\frac{du}{dt}=-k(m-\alpha+\beta u+T-hp) \quad \rightarrow (iii)$

The equation still contains two variables p and u solving equation (ii) for p

$\frac{du}{dt}=-k(m-p) \Rightarrow p=m+\frac{1}{k}\frac{du}{dt} \quad \rightarrow (iv)$

Put (iv) in (iii)

$\frac{du}{dt}=-k[m-\alpha+\beta u+T-h(m+\frac{1}{k}\frac{du}{dt})]$

$\frac{du}{dt}=-k[m-\alpha+T-hm]-k\beta u+h\frac{du}{dt}$

$(1-h)\frac{du}{dt}+k\beta u=-k[m-\alpha+T-hm]$

$\frac{du}{dt}+\frac{k\beta}{(1-h)}u=\frac{-k}{(1-h)}[m-\alpha+T-hm]$

$(Here \quad h\neq 1)$

Differential equation (1^{st} order) in variable u

(c) How do these equations different fundamentally from one we obtain under the adaptive expectation hypothesis?

Answer:

These equations are of first order while the equation we obtain under the adaptive expectation hypothesis is of 2^{nd} order

(d) What change in parameter restriction is now necessary to make the new differential equation meaningful?

Answer:

These new differential equations will be only and only meaningful if

$h\neq 1$

Q.4. Give the model:

$p=\frac{1}{6}-3u+\frac{1}{3}\pi \quad \rightarrow (i)$

$\frac{d\pi}{dt}=\frac{3}{4}(p-\pi) \quad \rightarrow (ii)$

$\frac{du}{dt}=-\frac{1}{2}(m-p) \quad \rightarrow (iii)$

Substituting (i) in (ii)

$\frac{d\pi}{dt}=\frac{3}{4}\left(\frac{1}{6}-3u+\frac{1}{3}\pi\right)-\frac{3}{4}\pi$

$\frac{d\pi}{dt}=\frac{1}{8}-\frac{9}{4}u+\frac{1}{4}\pi-\frac{3}{4}\pi$

$\frac{d\pi}{dt}=\frac{1}{8}-\frac{9}{4}u-\frac{2}{4}\pi$

Taking derivative W.r.t "t" to create place for $\frac{du}{dt}$

$\frac{d^2\pi}{dt^2}=-\frac{9}{4}\frac{du}{dt}-\frac{1}{2}\frac{d\pi}{dt}$

$\frac{d^2\pi}{dt^2}=-\frac{9}{4}\left[-\frac{1}{2}(m-p)\right]-\frac{1}{2}\frac{d\pi}{dt}$

$\frac{d^2\pi}{dt^2}=\frac{9}{8}m-\frac{9}{8}p-\frac{1}{2}\frac{d\pi}{dt} \quad \rightarrow (iv)$

Solving equation (ii) for p

$\frac{3}{4}p=\frac{d\pi}{dt}+\frac{3}{4}\pi$

$p=\frac{4}{3}\frac{d\pi}{dt}+\pi \quad \rightarrow (v)$

Substituting (v) in (iv)

$\frac{d^2\pi}{dt^2}=\frac{9}{8}m-\frac{9}{8}\left(\frac{4}{3}\frac{d\pi}{dt}+\pi\right)-\frac{1}{2}\frac{d\pi}{dt}$

$\frac{d^2\pi}{dt^2}=\frac{9}{8}m-\frac{3}{2}\frac{d\pi}{dt}-\frac{9}{8}\pi-\frac{1}{2}\frac{d\pi}{dt}$

$\frac{d^2\pi}{dt^2}=\frac{9}{8}m-\frac{3}{2}\frac{d\pi}{dt}-\frac{1}{2}\frac{d\pi}{dt}-\frac{9}{8}\pi$

$$\frac{d^2\pi}{dt^2}=\frac{9}{8}m-\frac{4}{2}\frac{d\pi}{dt}-\frac{9}{8}\pi$$

$$\frac{d^2\pi}{dt^2}=\frac{9}{8}m-2\frac{d\pi}{dt}-\frac{9}{8}\pi$$

$$\frac{d^2\pi}{dt^2}+2\frac{d\pi}{dt}+\frac{9}{8}\pi=\frac{9}{8}m$$

$$\pi_p=\frac{9m/8}{9/8}=m$$

For π_c

$$r^2+2r+\frac{9}{8}=0 \qquad a_1=2; \qquad a_2=\frac{9}{8}$$

$$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{1}{2}\left[-2\pm\sqrt{(2)^2-4(9/8)}\right]$$

$$r_1,\ r_2=\frac{1}{2}\left[-2\pm\sqrt{4-\frac{9}{2}}\right]=\frac{1}{2}\left[-2\pm\sqrt{\frac{1}{2}}\,i\right]$$

$$=\frac{-2}{2}\pm\frac{1}{\sqrt{2}}\frac{1}{2}i$$

$$r_1,\ r_2=-1\pm\frac{1}{2}\frac{1}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}}i=-1\pm\frac{\sqrt{2}}{4}i$$

$$h=-1 \qquad v=\frac{\sqrt{2}}{4}$$

$$\pi_c=e^{ht}(A_5\cos vt+A_6\sin vt)$$

$$\pi_c=e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)$$

$$\pi_t=\pi_c+\pi_p$$

$$\pi_t=e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)+m$$

As $t\to\infty \quad \pi_c\to 0 \quad \pi_t\to m$ and solution is dynamically stable and time path is convergent with damped fluctuations

$$p=\frac{4}{3}\frac{d\pi}{dt}+\pi$$

We require π and $\frac{d\pi}{dt}$

$$\frac{d\pi}{dt}=e^{-t}\left(-\frac{\sqrt{2}}{4}A_5\sin\frac{\sqrt{2}}{4}t+\frac{\sqrt{2}}{4}A_6\cos\frac{\sqrt{2}}{4}t\right)-e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)$$

$$p_{(t)}=\frac{4}{3}e^{-t}\left(-\frac{\sqrt{2}}{4}A_5\sin\frac{\sqrt{2}}{4}t+\frac{\sqrt{2}}{4}A_6\cos\frac{\sqrt{2}}{4}t\right)-\frac{4}{3}e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)+e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}+A_6\sin\frac{\sqrt{2}}{4}t\right)+m$$

$$p_{(t)}=\frac{4}{3}e^{-t}\left(-\frac{\sqrt{2}}{4}A_5\sin\frac{\sqrt{2}}{4}t+\frac{\sqrt{2}}{4}A_6\cos\frac{\sqrt{2}}{4}t\right)-\frac{1}{3}e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)+m$$

$$p_{(t)}=\frac{\sqrt{2}}{3}e^{-t}\left(-A_5\sin\frac{\sqrt{2}}{4}t+A_6\cos\frac{\sqrt{2}}{4}t\right)-\frac{1}{3}e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)+m$$

$$p_{(t)}=\frac{-1}{3}e^{-t}\left[(A_5-A_6\sqrt{2})\cos\frac{\sqrt{2}}{4}t+(A_6+A_5\sqrt{2})\sin\frac{\sqrt{2}}{4}t\right]+m$$

$$p_{(t)}=\frac{-1}{3}e^{-t}\left[A_1\cos\frac{\sqrt{2}}{4}t+A_2\sin\frac{\sqrt{2}}{4}t\right]+m$$

$$A_1=A_5-A_6\sqrt{2}$$
$$A_2=A_6+A_5\sqrt{2}$$

As $t\to\infty \quad p_c\to 0$ *and* $p_t\to m$ and solution is dynamically stable and time path is convergent with damped fluctuation

Finding the time path of u

$$3u=\frac{1}{6}-p+\frac{1}{3}\pi$$

$$u=\frac{1}{18}-\frac{1}{3}p+\frac{1}{9}\pi$$

$$u=\frac{1}{18}-\frac{1}{3}\left[-\frac{1}{3}e^{-t}\left(A_1\cos\frac{\sqrt{2}}{4}t+A_2\sin\frac{\sqrt{2}}{4}t\right]-\frac{1}{3}m\right.$$

$$+\frac{1}{9}\left[e^{-t}\left(A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right)\right]+\frac{1}{9}m$$

$$u_{(t)}=\frac{1}{18}+\frac{1}{9}e^{-t}\left(A_1\cos\frac{\sqrt{2}}{4}t+A_2\sin\frac{\sqrt{2}}{4}t\right)-\frac{1}{3}m$$

$$+\frac{1}{9}\left[e^{-t}A_5\cos\frac{\sqrt{2}}{4}t+A_6\sin\frac{\sqrt{2}}{4}t\right]+\frac{1}{9}m$$

$$u_{(t)}=\frac{1}{9}e^{-t}\left[(A_1+A_5)\cos\frac{\sqrt{2}}{4}t+(A_2+A_6)\sin\frac{\sqrt{2}}{4}t\right]$$

$$-\frac{1}{3}m+\frac{1}{9}m+\frac{1}{18}$$

$$u_{(t)}=\frac{1}{9}e^{-t}\left[B_1\cos\frac{\sqrt{2}}{4}t+B_2\sin\frac{\sqrt{2}}{4}t\right]-\frac{2}{9}m+\frac{1}{18}$$

As $t\to\infty \quad u_c\to 0 \quad and \quad u_t\to\frac{1}{18}-\frac{2}{9}m$

The solution is dynamically stable and time path is convergent with damped fluctuation.

(b) Are the time path still fluctuating still convergent?

Answer:

Yes the time path is still fluctuating and still convergent.

(c) What are $\bar{P}$ and $\bar{u}$ the inter temporal equilibrium values?

$\bar{p}=m$

$\bar{u}=-\frac{2}{9}m+\frac{1}{18}$

(d) Is it still true that $\bar{u}$ is functionally unrelated to $\bar{P}$? If we now link these two equilibrium values to each other in a long run Philips curve. Can we still get a vertical curve? What assumption in example 1 is thus crucial for deriving a long-run Philips curve?

Answer:

Here $\bar{U}$ and $\bar{P}$ are related to each other. Now we have a relation between equilibrium value of U and P.

$\bar{p}=m \quad and \quad \bar{u}=-\frac{2}{9}m+\frac{1}{18}$

The above equation show that even if $\bar{P}_{(0)}$ which is actual rate of inflation. There should be still unemployment which is natural rate of unemployment $\frac{1}{18}$ this equation shows inverse relationship between rate of inflation and rate of unemployment. So if we link these two equilibrium values to each other in long run Philips curve, we cannot get a vertical curve. The crucial assumption in example 1 is the value of h = 1. If we relax this assumption, we cannot derive a vertical long run Philips curve.

Higher-Order Linear Differential Equation with Constant Coefficient and Constant Term:

Suppose we are seeking the solution of nth order linear differential equation

$y_{(t)}n+a_1y_{(t)}^{n-1}+\ldots\ldots\ldots\ldots\ldots+a_{n-1}y'_{(t)}+a_ny=b$

Finding the Solution:

In this case of constant coefficient and constant term, the presence of higher derivative does not materially affect the method of finding the particular integral. If we try a simplest possible type of solution y = k we can see that all the derivatives from $y'_{(t)}$ to $y^n_{(t)}$ will be zero and above equation will reduce to

$a_nk=b$

$$y_p=k=\frac{b}{a_n} \qquad a_n\neq 0$$

If however $a_n = 0$, we must try a solution of the form $y=kt$ then $y'_{(t)}=k$ but all the higher derivatives will vanish and above equation can be reduced to

$$a_{n-1}k=b$$

$$k=\frac{b}{a_{n-1}}$$

$$y_p=kt=\frac{b}{a_{n-1}}t \qquad (a_n=0;\quad a_{n-1}\neq 0)$$

If it happens that $a_n=a_{n-1}=0$ then try a solution of the form

$$y=kt^2$$

Further adaptation of this procedure should be obvious, i.e. then

$$y'=2kt$$

$$y''_{(t)}=2k$$

But all the higher derivatives will vanish and above equation can be reduced to

$$a_{n-2}k=b$$

$$k=\frac{b}{a_{n-2}}$$

$$y_p=kt^2=\frac{b}{a_{n-2}}t^2 \quad (a_n=a_{n-1}=0;\quad a_{n-2}\neq 0)$$

Complementary Function:

As for the complementary function, inclusion of the higher derivative in the differential equation has the effect of raising the degree of the characteristic equation. The complementary function is defined as the general solution of the reduced equation.

$$y_{(t)}^{n}+a_1y_{(t)}^{(n-1)}+\dots\dots\dots\dots+a_{n-1}y'_{(t)}+a_ny=0$$

Trying a solution of this form

$$y=Ae^{rt} \qquad Ae^{rt}\neq 0$$

Then $y'=rAe^{rt}$

$$y''=r^2Ae^{rt}$$

$$\vdots \qquad \vdots$$

$$y^{n-1}=r^{n-1}Ae^{rt}$$

$$y^n=r^nAe^{rt}$$

$$r^nAe^{rt}+a_1r^{n-1}Ae^{rt}+\dots\dots\dots+a_{n-1}rAe^{rt}+a_nAe^{rt}=0$$

Hence we can rewrite the above reduced equation as

$$Ae^{rt}(r^n+a_1r^{n-1}+\dots\dots\dots+a_{n-1}r+a_n)=0$$

Dividing both sides by Ae^{rt}

$$r^n+a_1r^{n-1}+\dots\dots\dots+a_{n-1}r+a_n=0$$

There must be n roots to this polynomial. Thus our complementary function will be

$$y_c=A_1e^{rt}+A_2e^{rt}+\dots\dots\dots+A_ne^{rt}=\sum_{i=1}^{n}A_ie^{rit}$$

However some modification must be made incase n roots are not all real distinct. First suppose that there are repeated roots say

$r_1=r_2=r_3$ then to avoid "collapsing" we must write the first three terms of the solution as

$A_1e^{r_1t}+A_2te^{r_1t}+A_3t^2e^{r_1t}$ In case we have $r_4=r_1$ as well the fourth term must be alter to $A_4t^3e^{r_1t}$ etc.

Second suppose that two of the roots are complex say.

$$r_5,\ r_6=h\pm vi$$

Then the fifth and the sixth term in the above solution should be combined into the following expression.

$$e^{ht}(A_5\cos vt+A_6\sin vt)$$

The general solution will be the sum of the complementary function y_c and particular integral y_p.

$$y_t=y_c+y_p$$

To definitize the solution as many as n initial conditions will be required

Example 1:

Find the general solution of

$$y_{(t)}^{(4)}+6y_{(t)}'''+14y_{(t)}''+16y_{(t)}'+8y=24$$

$$a_1=6;\quad a_2=14;\quad a_3=16;\quad a_4=8;\quad b=24$$

$$y_p=\frac{b}{a_4}=\frac{24}{8}=3$$

Characteristic equation will be

$$r^4+6r^3+14r^2+16r+8=0$$

By using synthetic division

$$r=\pm1\pm2$$

$r^4+6r^3+14r^2+16r+18=0$
$(-2)^4+6(-2)^3+14(-2)^2+16(-2)+18=0$
$16-48+56-32+8=0$
$0=0$

Hence r = −2

−2	1	6	14	16	8
		−2	−8	−12	−8
	1	4	6	4	0

$(r+2)(r^3+4r^2+6r+4)=0$

$r^3+4r^2+6r+4=0 \qquad r=-2$

$-8+16-12+4=0 \qquad r+2=0$

$0=0$

−2	1	4	6	4
		−2	−4	−4
	1	2	2	0

$(r+2)(r+2)(r^2+2r+2)=0$

$r^2+2r+2=0$

$a_1=2; \quad a_2=2$

$$r_1,\ r_2=\frac{-2\pm\sqrt{(2)^2-4(2)}}{2}=\frac{-2\pm\sqrt{4-8}}{2}=\frac{-2\pm 2i}{2}$$

$r_1,\ r_2=-1\pm i$

Hence complementary function will be

$y_c=A_3e^{-2t}+A_4te^{-2t}+e^t(A_5\cos t+A_6\sin t)$

$y_t=y_c+y_p$

$y_t=A_3e^{-2t}+A_4te^{-2t}+e^{-t}(A_5\cos t+A_6\sin t)+3$

As $t\to\infty \quad y_c\to 0 \quad Hence \quad y_t\to 3$ thus solution will be dynamically stable

Convergence and Routh Theorem:

We can find a way of ascertaining the convergent and divergence of a time path without having to solve for the characteristic roots. Fortunately, there does exist such a method, which can provide a qualitative (though non graphic) analysis of a differential equation. This method is to be found in the Routh theorem which states that "The real parts of all the roots of the nth degree polynomial equation $a_0r^n+a_1r^{n-1}+\ldots\ldots\ldots+a_{n-1}r+a_n=0$ are negative if and only if the first n of the following sequence of determinants are all positive.

$$|a|;\begin{vmatrix}a_1 & a_3\\ a_0 & a_2\end{vmatrix};\begin{vmatrix}a_1 & a_3 & a_5\\ a_0 & a_2 & a_4\\ 0 & a_1 & a_3\end{vmatrix};\begin{vmatrix}a_1 & a_3 & a_5 & a_7\\ a_0 & a_2 & a_4 & a_6\\ 0 & a_1 & a_3 & a_5\\ 0 & a_0 & a_2 & a_4\end{vmatrix};\ldots\ldots\ldots\ldots$$

Exercise 15.7:

Q.1. Find the particular integral of the following:

(a) $y'''_{(t)}+2y''_{(t)}+y'_{(t)}+2y=8$

Try a solution of the form

$y=k$

Then $y'_{(t)}=y''_{(t)}=y'''_{(t)}=0$

Hence above equation will reduce to

$2k=8$

$k=\frac{8}{2}=4 \qquad y_p=4$

(b) $y'''_{(t)}+y''_{(t)}=+3y'_{(t)}=1$

Try a solution of the form

$y=kt \qquad as \quad a_3=0$

Then $y'_{(t)}=k$

And $y''_{(t)}=y'''_{(t)}=0$

Putting into above equation

$3k=1$

$k=\frac{1}{3} \qquad y_p=\frac{1}{3}$

(c) $3y'''_{(t)}+9y''_{(t)}=\frac{1}{3}$

As $a_2=a_3=0$

Hence try a solution of the form

$y_{(t)}kt^2$

$y'_{(t)}=2kt$

$y''_{(t)}=2k$

$y'''_{(t)}=0$

Substituting these values into above equation

$9y''_{(t)}=1$

$9(2)k=1 \Rightarrow 18k=1 \Rightarrow \frac{1}{18}$

$y_p=kt^2=\frac{1}{18}t^2$

(d) $y^{(4)}_{(t)}+y''_{(t)}=4$

Try $y=kt^2$ As $a_3=a_4=0$

$y'_{(t)}=2kt$

$y''_{(t)}=2k$

$y'''_{(t)}=y^{(4)}_{(t)}=0$

Putting these values into above equation

$2k=4$

$k=\frac{4}{2}=2$

$y_p=kt^2=2t^2$

Q.2. Find the y_p and the y_c (and hence the general solution) of:

(a) $y'''_{(t)}-2y''_{(t)}-y'_{(t)}+2y=4$

$y_p=\frac{b}{a_3}=\frac{4}{2}=2$

$r^3-2r^2-r+2=0$

Try r = 1

$1-2-1+2=0$

$0=0$

Hence r = 1

$r-1=0$

Applying factoring

$(r-1)(r^2-r-2)=0$

$a_1=-1;\quad a_2=-2$

$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{1\pm\sqrt{(-1)^2-4(-2)}}{2}$

$r_1,\ r_2=\frac{1\pm\sqrt{1+8}}{2}=\frac{1}{2}\pm\frac{3}{2}=2,\ -1$

$y_c=A_1e^t+A_2e^{2t}+A_3e^{-t}$

$y_{(t)}=y_c+y_p$

$y_{(t)}=A_1e^t+A_2e^{2t}+A_3e^{-t}+2$

(b) $y'''_{(t)}+7y''_{(t)}+15\{y'_{(t)}+9y=0$¿

Since b = 0

Therefore $y_p=0$

Forming the characteristic equation

$r^3+7r^2+15r+9=0$

Try r = −1

$-1+7-15+9=0$

$16-16=0$

$0=0$

Hence r = −1

$r+1=0$

−1	1	7	15	9
		−1	−6	−9
	1	6	9	0

$(r+1)(r^2+6r+9)=0$

$a_1=6;a_2=9$

$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-6\pm\sqrt{(6)^2-4(9)}}{2}$

$r_1,\ r_2=\frac{-6}{2}=-3$

Hence complementary function will be

$y_c=A_1e^{-t}+A_2e^{-3t}+A_3te^{-3t}$

$y_{(t)}=y_c+y_p$

$y_{(t)}=A_1e^{-t}+A_2e^{-3t}+A_3te^{-3t}$

As $t\rightarrow\infty \quad y_c\rightarrow 0$ hence solution is dynamically stable

(c) $y'''_{(t)}+6y''_{(t)}+10\{y'_{(t)}+8y=8$¿

$a_1=6\ ;\ a_2=10\ ;\ a_3=8\ ;\ b=8$

$y_p=\frac{b}{a_3}=\frac{8}{8}=1$

For y_c

$r^3+6r^2+10r+8=0$

Try r = −4

−4	1	6	10	8
		−4	−8	−8
	1	2	2	0

$(r+4)(r^2+2r+2)=0$

$a_1=2\ ;\ a_2=2$

$$r_1,\ r_2=\frac{-a_1\pm\sqrt{a_1^2-4a_2}}{2}=\frac{-2\pm\sqrt{(2)^2-4(2)}}{2}$$

$$r_1,\ r_2=\frac{-2\pm 2i}{2}=-1\pm i$$

$h=-1 \quad v=1$

Hence complementary function will be

$y_c=A_1e^{-4t}+e^{-t}(A_5\cos t+A_6\sin t)$

$y_t=y_c+y_p$

$y_t=Ae^{-4t}+e^{-t}(A_5\cos t+A_6\sin t)+1$

Q.3 On the basis of the signs of the characteristic roots obtained in the preceding problem analyzes the dynamic stability of equilibrium. Then check your answer by the Routh theorem.

(a) As $t\to\infty$ e^t and $e^{2t}\to\infty$ hence equilibrium is dynamically unstable.

By Routh theorem

$|a_1|=|-2|<0$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}=\begin{vmatrix} -2 & 2 \\ 1 & -1 \end{vmatrix}=2-2=0$$

Since all the determinants are not positive. Therefore time path will be divergent and solution will be dynamically unstable.

(b) As $t\to\infty$ $y_c\to 0$ hence solution is dynamically stable

By Routh theorem

$|a_1|=|7|=7>0$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}=\begin{vmatrix} 7 & 9 \\ 1 & 15 \end{vmatrix}=105-9=96>0$$

Since all the determinants are positive. The time path will be convergent and dynamically the equilibrium will be stable.

(c) As $t\to\infty$ $y_c\to 0$ *and* $y_t\to 1$ hence equilibrium is dynamically stable.

By Routh theorem

$|a_1|=|6|>0$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}=\begin{vmatrix} 6 & 8 \\ 1 & 10 \end{vmatrix}=60-8=58>0$$

Since the determinants are positive therefore solution is dynamically stable and time path will be convergent.

Q.4. Without finding the characteristic roots determines whether the following differential equation will give rise to convergent time paths.

(a) $y'''_{(t)}-10\{y''_{(t)}+27\{y¿'_{(t)}-18y=3¿$

$|a_1|=-10<0$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}=\begin{vmatrix} -10 & -18 \\ 1 & 27 \end{vmatrix}=-270+18=-252<0$$

$$\begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix}=\begin{vmatrix} -10 & -18 & 0 \\ 1 & 27 & 0 \\ 0 & -10 & -18 \end{vmatrix}$$

$$=-10\begin{vmatrix} 27 & 0 \\ -10 & -18 \end{vmatrix}+18\begin{vmatrix} 1 & 0 \\ 0 & -18 \end{vmatrix}-0$$

$=-5184<0$

Since all the determinants are negative, the time path will be divergent.

(b) $y'''_{(t)}+11\{y''_{(t)}+34\{y¿'_{(t)}+24y=5¿$

$a_1=11 \quad a_2=34 \quad a_3=24$

$|a_1|=11>0$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}=\begin{vmatrix} 11 & 24 \\ 1 & 34 \end{vmatrix}=374-24=250>0$$

Since all the determinants are positive. Therefore time path will be convergent.

(c) $y'''_{(t)}+4y''_{(t)}+5y'_{(t)}-2y=-2$

$a_1=4 \quad a_2=5 \quad a_3=-2 \quad a_0=1$

$|a_1|=4>0$

$$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}=\begin{vmatrix} 4 & -2 \\ 1 & 5 \end{vmatrix}=20+2=22>0$$

$$\begin{vmatrix} a_1 & a_3 & a_5 \\ a_0 & a_2 & a_4 \\ 0 & a_1 & a_3 \end{vmatrix} = \begin{vmatrix} 4 & -2 & 0 \\ 1 & 5 & 0 \\ 0 & 4 & -2 \end{vmatrix}$$

$$=4\begin{vmatrix} 5 & 0 \\ 4 & -2 \end{vmatrix} + 2\begin{vmatrix} 1 & 0 \\ 0 & -2 \end{vmatrix} = -40-4=-44<0$$

Since all the determinants are not positive. Therefore time path is divergent.

Q.5. Deduce from the Routh theorem that for the second−order differential equation

$y''_{(t)}+a_1 y'_{(t)}+a_2 y=b$

The solution path will be convergent regardless of initial conditions if and only if the coefficients a_1 and a_2 are both positive.

Answer:

For $|a_1|>0$

$\begin{vmatrix} a_1 & a_3 \\ a_0 & a_2 \end{vmatrix}>0$ It is necessary that $a_1>0$, $a_1 a_2>0$

Given $a_1>0$

$a_1 a_2$, will be positive if and only if $a_2>0$